

Control Architecture Design Document

PCS3440 + BESS Microgrid — Three-Tier Hierarchical Control

Project: PCS3440 + DELTERR Mi2 BESS Microgrid Controller Development

Date: June 27, 2026

System: 2 × 5 MW / 10 MWh BESS (10 MW / 20 MWh aggregate)

Platform: PSCAD/EMTDC v5.0.2

Revision: 1.0

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1. Architecture Overview

1.1 Design Philosophy

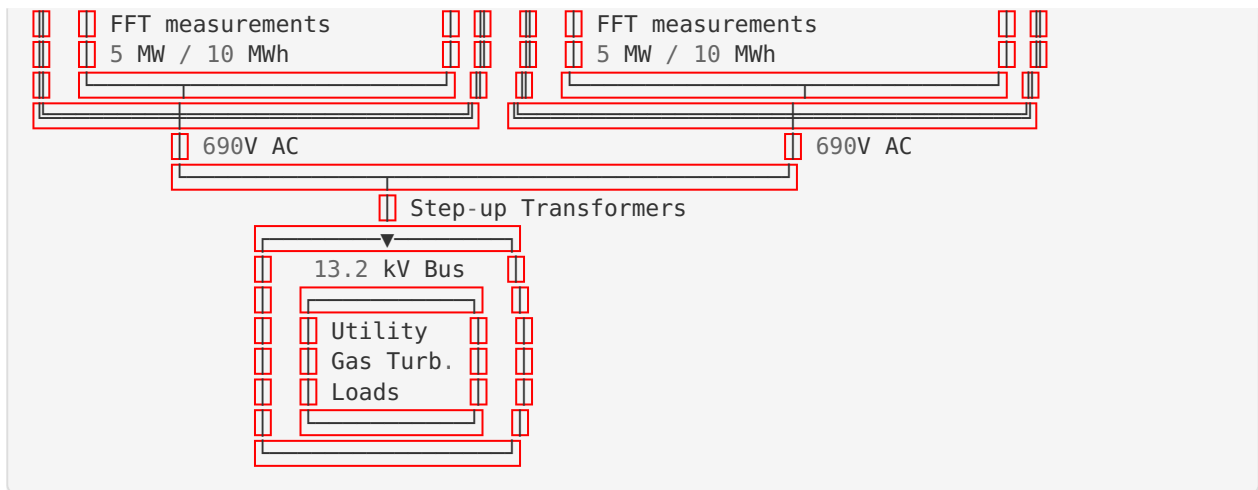
The control architecture is structured as three hierarchical layers that wrap the existing PCS3440 black-box controller (`Ctrl_1_2`). Each layer operates at a distinct time scale and abstraction level. **No modifications are made to the PCS internal firmware** — all new logic interacts exclusively through labeled ports, `PQref` inputs, and AuxCtrl slider switches.

1.2 Layer Summary

Layer	Name	Instances	Primary Function	Update Rate	Time Step
A	Unit Controller (UC)	2 (one per PCS)	Safety envelope, ramp limiting, SOC clamping	1 ms	50 μ s – 1 ms
B	Microgrid Controller (MGC)	1	Multi-unit coordination, state machine, islanding	10 ms	1 – 10 ms
C	SCADA / EMS Emulator	1	Plant-level dispatch, scheduling, optimization	1 s	100 ms – 1 s

1.3 Master Architecture Diagram





2. Layer A — Unit Controller (UC)

2.1 Purpose and Scope

Each Unit Controller (UC) provides a **safety envelope** around one PCS+BESS black box. It:

- Translates MGC commands into PCS-compatible `PQref` setpoints
- Enforces ramp-rate limits to prevent grid disturbances
- Clamps power commands based on SOC boundaries (10%–95%)
- Latches faults and forces safe states
- Manages operating mode transitions (PQ ↔ GFM)
- Provides status feedback to the MGC

The UC does **not** implement current control, PLL, or PWM — those are handled internally by `Ctrl_1_2`.

2.2 Input Signal Table

2.2.1 Inputs from PCS+BESS Subsystem (BESS_1_1)

#	Signal Name	PSCAD Label	Type	Dim	Units	Range	Source	Description
I-A01	P_meas	P	Real	1	kW	-5000 to +5000	FFT_Mod_2	Measured active power (+ = discharge)
I-A02	Q_meas	Q	Real	1	kVAR	-5000 to +5000	FFT_Mod_2	Measured reactive power
I-A03	SOC	BatSOC	Real	1	%	0 – 100	BESS model	Battery state of charge
I-A04	V_bat	Bat-volt	Real	1	Vdc	0 – 1500	BESS model	Battery terminal voltage
I-A05	V_dc	Vdc_meas	Real	1	Vdc	0 – 1500	DC link cap	DC bus voltage
I-A06	V_ac_rms[3]	Vlrms	Real	3	V	0 – 1000	FFT_Mod_2	Line voltage RMS (Vab, Vbc, Vca)
I-A07	PCS_Fault[4]	PCS Fault	Int	4	bitmask	0x0000 – 0xFFFF	Ctrl_1_2	Fault status bitmask (4 channels)
I-A08	MCB_status	MCB	Int	1	bool	0/1	Breaker	Main circuit breaker state (1=closed)

#	Signal Name	PSCAD Label	Type	Dim	Units	Range	Source	Description
I-A09	I_active	I_active	Real	1	A	-5000 to +5000	FFT_Mod_2	Active current component
I-A10	I_reactive	I_reactive	Real	1	A	-5000 to +5000	FFT_Mod_2	Reactive current component
I-A11	MC_status[4]	MC	Int	4	bool	0/1	Con-tactors	Main contactor states
I-A12	DC_EN_status	DC_EN	Int	1	bool	0/1	DC system	DC system enabled flag
I-A13	Mode_PCS	Mode	Int	1	enum	0-4	Ctrl_1_2	Internal PCS operating mode

2.2.2 Inputs from Microgrid Controller (Layer B → Layer A)

#	Signal Name	Type	Dim	Units	Range	Update Rate	Description
I-A20	P_cmd	Real	1	kW	-5000 to +5000	10 ms	Active power command (+ = discharge)
I-A21	Q_cmd	Real	1	kVAR	-5000 to +5000	10 ms	Reactive power command
I-A22	Mode_cmd	Int	1	enum	0-4	10 ms	Mode command (see §2.5.4)
I-A23	Enable	Int	1	bool	0/1	10 ms	Master enable (0=disabled, 1=enabled)
I-A24	Ramp_limit	Real	1	kW/s	0 – 50000	10 ms	Dynamic ramp rate limit override
I-A25	V_dc_ref	Real	1	Vdc	1200 – 1450	10 ms	DC link voltage reference (nominal 1332)
I-A26	MGC_heartbeat	Int	1	counter	0 – 65535	10 ms	Incrementing heartbeat from MGC

2.3 Output Signal Table

2.3.1 Outputs to PCS+BESS Subsystem (UC → BESS_1_1)

#	Signal Name	PSCAD Port/Slider	Type	Dim	Units	Range	Description
O-A01	P_ref	PQref(1)	Real	1	kW	-5000 to +5000	Processed active power setpoint
O-A02	Q_ref	PQref(2)	Real	1	kVAR	-5000 to +5000	Processed reactive power setpoint
O-A03	RampRate_set	RampRate slider	Real	1	ms	10 – 1000	PCS internal ramp rate setting
O-A04	AVR_set	AVR slider	Int	1	bool	0/1	Q-V droop enable
O-A05	FW_EN_set	FW_EN slider	Int	1	bool	0/1	Frequency-watt enable
O-A06	VRT_Enable_set	VRT_Enable slider	Int	1	bool	0/1	Voltage ride-through enable
O-A07	PROT_Enable_set	PROT_Enable slider	Int	1	bool	0/1	V/F trip protection enable
O-A08	HPF_Enable_set	HPF_Enable slider	Int	1	bool	0/1	Grid voltage HPF enable
O-A09	VQ_V[4]	VQ_V1..V4 sliders	Real	4	0.1%	900–1200	Q-V droop

#	Signal Name	PSCAD Port/Slider	Type	Dim	Units	Range	Description
							voltage points
O-A10	VQ_Q[4]	VQ_Q1..Q4 sliders	Real	4	0.1%	-1000-1000	Q-V droop Q points
O-A11	MCB_DRV	MCB_DRV1	Int	1	bool	0/1	AC breaker close command

2.3.2 Outputs to Microgrid Controller (Layer A → Layer B)

#	Signal Name	Type	Dim	Units	Range	Update Rate	Description
O-A20	P_meas_out	Real	1	kW	-5000 to +5000	10 ms	Measured active power (pass-through)
O-A21	Q_meas_out	Real	1	kVAR	-5000 to +5000	10 ms	Measured reactive power (pass-through)
O-A22	SOC_out	Real	1	%	0 – 100	10 ms	Battery SOC (pass-through)
O-A23	V_ac_rms_out	Real	1	V	0 – 1000	10 ms	Average line voltage RMS
O-A24	V_dc_out	Real	1	Vdc	0 – 1500	10 ms	DC bus voltage (pass-through)
O-A25	UC_Status	Int	1	bitmask	0x0000–0xFFFF	10 ms	Status word (see §2.3.3)
O-A26	UC_Fault	Int	1	bitmask	0x0000–0xFFFF	10 ms	Fault word (see §2.3.4)
O-A27	UC_Mode	Int	1	enum	0–4	10 ms	Current operating mode
O-A28	P_available	Real	1	kW	0 – 5000	10 ms	Available discharge

#	Signal Name	Type	Dim	Units	Range	Update Rate	Description
							power (SOC-based)
O-A29	P_charge_avail	Real	1	kW	0 – 5000	10 ms	Available charge power (SOC-based)
O-A30	UC_heart beat	Int	1	counter	0 – 65535	10 ms	Incrementing heart-beat

2.3.3 UC_Status Bit Definitions

Bit	Name	Description
0	READY	UC initialized and ready to accept commands
1	ENABLED	Master enable is active
2	RUNNING	Power conversion is active (P_ref $\neq$ 0 or Q_ref $\neq$ 0)
3	MCB_CLOSED	AC main circuit breaker is closed
4	DC_OK	DC bus voltage within $\pm 5\%$ of nominal
5	SOC_NORMAL	SOC is within 15%–90% (not near limits)
6	SOC_LOW_WARN	SOC is between 10%–15% (approaching low limit)
7	SOC_HIGH_WARN	SOC is between 90%–95% (approaching high limit)
8	RAMP_ACTIVE	Output is currently ramp-rate limited
9	CLAMP_ACTIVE	Output is currently SOC-clamped
10	GFM_MODE	Grid-forming mode active
11	PQ_MODE	Grid-following (PQ) mode active
12	VRT_ACTIVE	Voltage ride-through event in progress
13–15	Reserved	—

2.3.4 UC_Fault Bit Definitions

Bit	Name	Description	Action
0	PCS_HW_FAULT	PCS hardware fault (from PCS_Fault bit-mask)	Force P=Q=0, open MCB
1	OC_TRIP	Overcurrent trip detected	Force P=Q=0, open MCB
2	DC_OV	DC bus overvoltage (>1450 V)	Reduce charge power
3	DC_UV	DC bus undervoltage (<1200 V)	Reduce discharge power
4	SOC_FLOOR	$SOC \leq 10\%$ — discharge blocked	Block discharge
5	SOC_CEILING	$SOC \geq 95\%$ — charge blocked	Block charge
6	COMM_LOSS	MGC heartbeat lost (>500 ms)	Ramp to zero, hold
7	AC_UV	AC undervoltage (<0.88 pu for >2s)	Trip per IEEE 1547
8	AC_OV	AC overvoltage (>1.10 pu for >1s)	Trip per IEEE 1547
9	TEMP_ALARM	Temperature alarm (reserved)	Derate power
10-15	Reserved	—	—

2.4 UC Internal Registers

Register	Type	Init	Description
P_ref_internal	Real	0.0	Ramp-filtered active power reference
Q_ref_internal	Real	0.0	Ramp-filtered reactive power reference
P_target	Real	0.0	SOC-clamped power target
fault_latch	Int	0x0000	Latched fault word
uc_state	Enum	IDLE	Current state machine state
heartbeat_timer	Real	0.0	Time since last MGC heartbeat
heartbeat_counter	Int	0	UC heartbeat counter
ramp_timer	Real	0.0	Ramp limiter time accumulator
fault_clear_timer	Real	0.0	Fault clear delay timer
last_mgc_hb	Int	0	Previous MGC heartbeat value

2.5 UC Control Logic

2.5.1 Ramp-Rate Limiter

The UC enforces a maximum rate of change on the P and Q references to prevent grid disturbance and comply with the 20 MW/min utility interconnection requirement.

Algorithm:

```

CONSTANTS:
  DEFAULT_RAMP = 833.33 kW/s    (= 50,000 kW/min = 5 MW in 6 s for single unit)
  MIN_RAMP     = 100 kW/s      (minimum ramp rate)
  MAX_RAMP     = 50,000 kW/s   (maximum ramp rate ☐ effectively instant)
  Ts           = 0.001 s      (1 ms sample period)

INPUTS:
  P_cmd        ☐ from MGC (desired setpoint)
  Ramp_limit   ☐ from MGC (dynamic override, 0 = use default)

OUTPUTS:
  P_ref_internal ☐ ramped active power reference

LOGIC:
  IF Ramp_limit > 0 THEN
    ramp_rate = CLAMP(Ramp_limit, MIN_RAMP, MAX_RAMP)
  ELSE
    ramp_rate = DEFAULT_RAMP
  END IF

  delta_max = ramp_rate ☒ Ts // max change per sample
  delta = P_cmd ☐ P_ref_internal

  IF |delta| ☐ delta_max THEN // target reached
    P_ref_internal = P_cmd
    RAMP_ACTIVE = 0
  ELSE
    P_ref_internal = P_ref_internal + SIGN(delta) ☒ delta_max
    RAMP_ACTIVE = 1
  END IF

```

The same algorithm applies independently to `Q_ref_internal` with `Q_cmd`.

Compliance check:

At the default 833.33 kW/s per unit, two units ramping together produce 1,666.67 kW/s = 100 MW/min. The MGC must set `Ramp_limit` appropriately such that the aggregate plant ramp stays within 20 MW/min (= 333.33 kW/s total = 166.67 kW/s per unit).

2.5.2 SOC Protection Logic

The SOC protection implements hard limits at 10% (floor) and 95% (ceiling) with linear derating in the warning zones.

Algorithm:

```

CONSTANTS:
SOC_FLOOR           = 10.0 % // Discharge blocked below this
SOC_CEILING         = 95.0 % // Charge blocked above this
SOC_LOW_WARN        = 15.0 % // Derating begins (discharge)
SOC_HIGH_WARN       = 90.0 % // Derating begins (charge)
P_RATED             = 5000 kW // Rated power

INPUTS:
SOC                 - battery state of charge (%)
P_target_raw        - from ramp limiter output (kW)

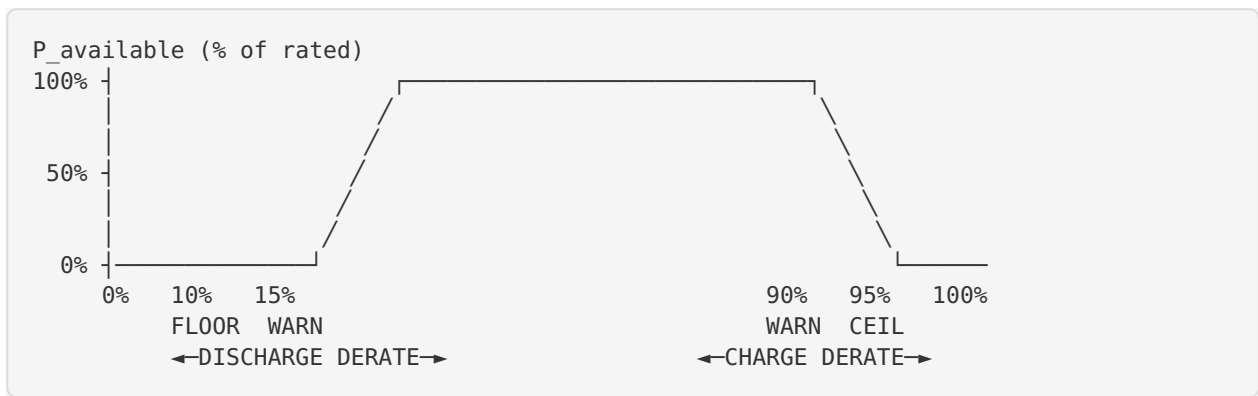
OUTPUTS:
P_target            - SOC-clamped power setpoint (kW)
CLAMP_ACTIVE        - flag

LOGIC:
// Discharge limiting (P_target_raw > 0 means discharge)
IF P_target_raw > 0 THEN
  IF SOC ≤ SOC_FLOOR THEN
    P_target = 0 // Hard stop
    SOC_FLOOR_fault = 1
    CLAMP_ACTIVE = 1
  ELSE IF SOC ≤ SOC_LOW_WARN THEN
    // Linear derating: 0% at SOC_FLOOR → 100% at SOC_LOW_WARN
    derate = (SOC - SOC_FLOOR) / (SOC_LOW_WARN - SOC_FLOOR)
    P_target = P_target_raw × derate
    CLAMP_ACTIVE = 1
  ELSE
    P_target = P_target_raw
    CLAMP_ACTIVE = 0
  END IF

  // Charge limiting (P_target_raw ≤ 0 means charge)
ELSE IF P_target_raw ≤ 0 THEN
  IF SOC ≥ SOC_CEILING THEN
    P_target = 0 // Hard stop
    SOC_CEILING_fault = 1
    CLAMP_ACTIVE = 1
  ELSE IF SOC ≥ SOC_HIGH_WARN THEN
    // Linear derating: 100% at SOC_HIGH_WARN → 0% at SOC_CEILING
    derate = (SOC_CEILING - SOC) / (SOC_CEILING - SOC_HIGH_WARN)
    P_target = P_target_raw × derate
    CLAMP_ACTIVE = 1
  ELSE
    P_target = P_target_raw
    CLAMP_ACTIVE = 0
  END IF
ELSE
  P_target = 0
  CLAMP_ACTIVE = 0
END IF

```

SOC Derating Curve:



2.5.3 Fault Handling Logic

```

CONSTANTS:
    FAULT_CLEAR_DELAY = 5.0 s    // Time to wait before clearing latched fault
    HB_TIMEOUT         = 0.5 s    // MGC heartbeat timeout

LOGIC (executed every 1 ms):

// ☐ PCS Hardware Fault ☐
IF ANY(PCS_Fault[0..3] ☒ 0) THEN
    fault_latch |= BIT(0)          // PCS_HW_FAULT
    P_ref = 0; Q_ref = 0
    MCB_DRV = 0                    // Open breaker

// ☐ DC Bus Overvoltage ☐
IF V_dc > 1450 THEN
    fault_latch |= BIT(2)          // DC_OV
    // Allow discharge only (to bring Vdc down)
    IF P_ref_internal < 0 THEN P_ref_internal = 0

// ☐ DC Bus Undervoltage ☐
IF V_dc < 1200 THEN
    fault_latch |= BIT(3)          // DC_UV
    // Allow charge only (to bring Vdc up)
    IF P_ref_internal > 0 THEN P_ref_internal = 0

// ☐ Communication Loss ☐
IF MGC_heartbeat == last_mgc_hb THEN
    heartbeat_timer += Ts
    IF heartbeat_timer > HB_TIMEOUT THEN
        fault_latch |= BIT(6)      // COMM_LOSS
        // Ramp to zero (don't slam)
        P_cmd_effective = 0
        Q_cmd_effective = 0
    ELSE
        heartbeat_timer = 0
        last_mgc_hb = MGC_heartbeat
        fault_latch &= ~BIT(6)     // Clear comm loss on heartbeat resume

// ☐ Fault Clear Logic ☐
// Faults 0, 1 require explicit clear command from MGC
// Faults 2-5 auto-clear when condition resolves
IF V_dc ☒ 1400 THEN fault_latch &= ~BIT(2)
IF V_dc ☒ 1250 THEN fault_latch &= ~BIT(3)
IF SOC > SOC_FLOOR THEN fault_latch &= ~BIT(4)
IF SOC < SOC_CEILING THEN fault_latch &= ~BIT(5)

// ☐ Master Fault Check ☐
IF fault_latch & (BIT(0) | BIT(1)) THEN
    // Hard faults ☐ force shutdown
    P_ref = 0; Q_ref = 0; MCB_DRV = 0
END IF

```

2.5.4 Mode Command Definitions

Mode_cm d	Name	AVR	FW_EN	VRT	PROT	Descrip- tion
0	IDLE	0	0	0	0	Standby — P=Q=0, MCB open
1	GFL (Grid- Following)	0	0	1	1	PQ mode, follow MGC P/Q com- mands
2	GFM (Grid- Forming)	1	1	1	1	Droop mode, V/f regulation
3	CHARGE	0	0	1	1	Charge- only mode (P_ref ≤ 0)
4	FAULT	0	0	0	0	Fault state — P=Q=0, awaiting clear

Mode Transition Logic:

```

ON Mode_cmd CHANGE:
  CASE Mode_cmd OF
    0 (IDLE):
      Set AVR=0, FW_EN=0, VRT_Enable=0, PROT_Enable=0
      Ramp P_ref  $\rightarrow$  0, Q_ref  $\rightarrow$  0
      After P_ref==0: Open MCB

    1 (GFL):
      Set AVR=0, FW_EN=0, VRT_Enable=1, PROT_Enable=1
      Close MCB
      After MCB confirmed closed: accept P_cmd, Q_cmd

    2 (GFM):
      Set AVR=1, FW_EN=1, VRT_Enable=1, PROT_Enable=1
      Load Q-V droop curve (VQ_V[4], VQ_Q[4])
      Close MCB
      // P follows droop + P_cmd bias
      // Q follows droop curve

    3 (CHARGE):
      Set AVR=0, FW_EN=0, VRT_Enable=1, PROT_Enable=1
      Close MCB
      // Only negative P_cmd accepted (charge)
      IF P_cmd > 0 THEN P_cmd_effective = 0

    4 (FAULT):
      Ramp P_ref  $\rightarrow$  0, Q_ref  $\rightarrow$  0
      After P_ref==0: Open MCB
      Set AVR=0, FW_EN=0, VRT_Enable=0, PROT_Enable=0
  END CASE

```

2.5.5 Available Power Calculation

The UC continuously reports how much power it can source or sink, which the MGC uses for power sharing:

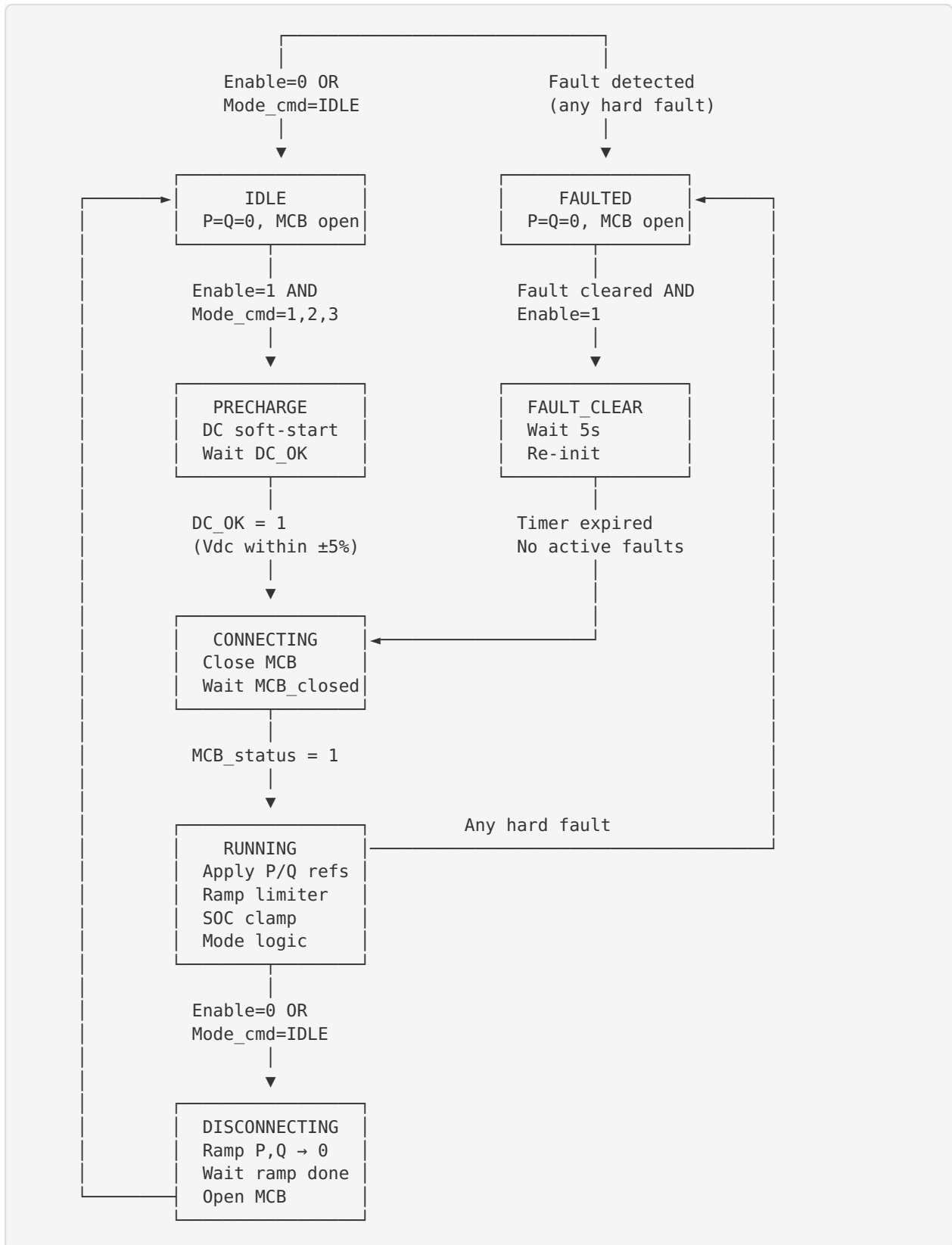
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// Discharge availability
IF SOC > SOC_LOW_WARN THEN
  P_available = P_RATED
ELSE IF SOC > SOC_FLOOR THEN
  P_available = P_RATED × (SOC - SOC_FLOOR) / (SOC_LOW_WARN - SOC_FLOOR)
ELSE
  P_available = 0

// Charge availability
IF SOC < SOC_HIGH_WARN THEN
  P_charge_avail = P_RATED
ELSE IF SOC < SOC_CEILING THEN
  P_charge_avail = P_RATED × (SOC_CEILING - SOC) / (SOC_CEILING - SOC_HIGH_WARN)
ELSE
  P_charge_avail = 0

```

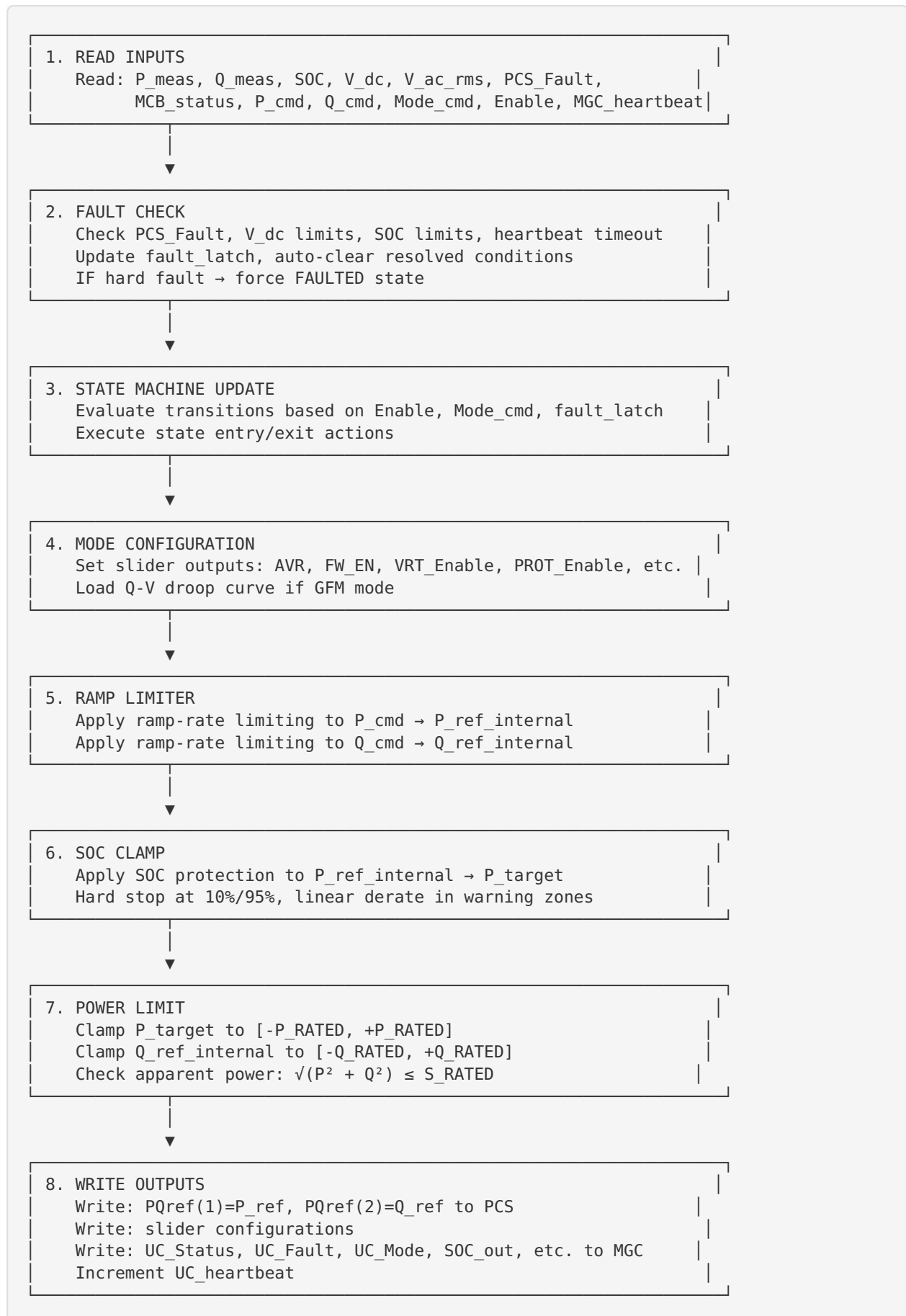

2.6 UC State Machine



State Descriptions:

State	Entry Condition	Actions	Exit Conditions
IDLE	Power-on default, or explicit disable	$P=Q=0$, MCB open, all outputs zeroed	Enable=1 AND valid Mode_cmd
PRECHARGE	From IDLE when enabled	Assert DCC_SS_EN, monitor Vdc	DC_OK (Vdc within $\pm 5\%$ of 1332V)
CONNECTING	DC bus ready	Close MCB, configure mode sliders	MCB_status=1 confirmed
RUNNING	MCB closed	Execute ramp limiter, SOC clamp, apply P/Q refs	Enable=0, Mode=IDLE, or Fault refs
DISCONNECTING	Disable or shutdown	Ramp P,Q $\rightarrow 0$, then open MCB	$P \approx 0$ AND MCB open $\rightarrow$ IDLE
FAULTED	Hard fault detected	Force $P=Q=0$, open MCB, latch fault	Fault cleared + 5s delay
FAULT_CLEAR	Fault condition resolved	Wait 5s cooldown, re-initialize	Timer expired $\rightarrow$ CONNECTING

2.7 UC Processing Pipeline (per 1 ms cycle)



3. Layer B — Microgrid Controller (MGC)

3.1 Purpose and Scope

The Microgrid Controller is a single, centralized coordinator that:

- Distributes plant-level power commands across multiple Unit Controllers using SOC-weighted sharing
- Manages the microgrid state machine (grid-tied, islanded, black-start, etc.)
- Implements islanding detection and resynchronization
- Controls load shedding and priority load management
- Coordinates with gas turbine and utility breaker
- Reports aggregated status to the SCADA/EMS layer

3.2 Input Signal Table

3.2.1 Inputs from Unit Controllers (×2)

#	Signal Name	Type	Dim	Units	Range	Source	Description
I-B01	UC1_P_meas	Real	1	kW	-5000 to +5000	UC1	Unit 1 measured active power
I-B02	UC1_Q_meas	Real	1	kVAR	-5000 to +5000	UC1	Unit 1 measured reactive power
I-B03	UC1_SOC	Real	1	%	0 – 100	UC1	Unit 1 battery SOC
I-B04	UC1_V_ac_rms	Real	1	V	0 – 1000	UC1	Unit 1 AC voltage RMS
I-B05	UC1_V_dc	Real	1	Vdc	0 – 1500	UC1	Unit 1 DC bus voltage
I-B06	UC1_Status	Int	1	bitmask	0x0000–0xFFFF	UC1	Unit 1 status word
I-B07	UC1_Fault	Int	1	bitmask	0x0000–0xFFFF	UC1	Unit 1 fault word
I-B08	UC1_Mode	Int	1	enum	0–4	UC1	Unit 1 operating mode
I-B09	UC1_P_avail	Real	1	kW	0 – 5000	UC1	Unit 1 available discharge power
I-B10	UC1_P_charging_avail	Real	1	kW	0 – 5000	UC1	Unit 1 available

#	Signal Name	Type	Dim	Units	Range	Source	Description
							charge power
I-B11	UC1_heartbeat	Int	1	counter	0-65535	UC1	Unit 1 heart-beat
I-B12..B22	UC2_*	—	—	—	—	UC2	(Identical signals from Unit Controller 2)

3.2.2 Inputs from Grid / POI Sensors

#	Signal Name	Type	Dim	Units	Range	Source	Description
I-B30	V_poi	Real	1	V	0 – 15000	POI meter	Point of interconnection voltage
I-B31	P_poi	Real	1	kW	-15000 to +15000	POI meter	POI active power (+ = export)
I-B32	Q_poi	Real	1	kVAR	-15000 to +15000	POI meter	POI reactive power
I-B33	f_grid	Real	1	Hz	55 – 65	PLL / frequency meter	Grid frequency
I-B34	V_grid_abcd[3]	Real	3	V	±1000	Grid sensors	3-phase grid voltages
I-B35	df_dt	Real	1	Hz/s	-10 to +10	Calculated	Rate of change of frequency (ROCOF)
I-B36	V_phase_angle	Real	1	deg	-180 to +180	PLL	Grid voltage phase angle
I-B37	Utility_BRK	Int	1	bool	0/1	Breaker	Utility breaker status (1=closed)

3.2.3 Inputs from Gas Turbine

#	Signal Name	Type	Dim	Units	Range	Source	Description
I-B40	GT_P_meas	Real	1	kW	0 – 500	GT meter	Gas turbine active power
I-B41	GT_V_meas	Real	1	V	0 – 15000	GT meter	Gas turbine terminal voltage
I-B42	GT_f_meas	Real	1	Hz	55 – 65	GT meter	Gas turbine frequency
I-B43	GT_Status	Int	1	enum	0-3	GT ctrl	0=Off, 1=Starting, 2=Running, 3=Fault
I-B44	GT_Ready	Int	1	bool	0/1	GT ctrl	GT ready to accept load

3.2.4 Inputs from Building Load

#	Signal Name	Type	Dim	Units	Range	Source	Description
I-B50	Load_P_total	Real	1	kW	0 – 10000	Load meter	Total building load (active)
I-B51	Load_Q_total	Real	1	kVAR	0 – 10000	Load meter	Total building load (reactive)
I-B52	Load_Priority[4]	Real	4	kW	varies	Load meters	Per-priority-tier load measurements

3.2.5 Inputs from EMS / SCADA (Layer C → Layer B)

#	Signal Name	Type	Dim	Units	Range	Update Rate	Description
I-B60	P_dispatch	Real	1	kW	-10000 to +10000	1 s	Plant-level P dispatch command
I-B61	Q_dispatch	Real	1	kVAR	-10000 to +10000	1 s	Plant-level Q dispatch command
I-B62	Plant-Mode	Int	1	enum	0-5	1 s	Plant operating mode (see §3.5.6)
I-B63	DR_flag	Int	1	bool	0/1	1 s	Demand response event active
I-B64	EMS_heartbeat	Int	1	counter	0-65535	1 s	EMS heart-beat counter

3.3 Output Signal Table

3.3.1 Outputs to Unit Controllers (×2)

#	Signal Name	Type	Dim	Units	Range	Update Rate	Description
O-B01	UC1_P_cmd	Real	1	kW	-5000 to +5000	10 ms	Unit 1 active power command
O-B02	UC1_Q_cmd	Real	1	kVAR	-5000 to +5000	10 ms	Unit 1 reactive power command
O-B03	UC1_Mode_cmd	Int	1	enum	0-4	10 ms	Unit 1 mode command
O-B04	UC1_Enable	Int	1	bool	0/1	10 ms	Unit 1 master enable
O-B05	UC1_Ramp_limit	Real	1	kW/s	0-50000	10 ms	Unit 1 ramp rate override
O-B06	UC1_V_dc_ref	Real	1	Vdc	1200-1450	10 ms	Unit 1 DC voltage reference
O-B07	UC1_MGC_heartbeat	Int	1	counter	0-65535	10 ms	Heart-beat to UC1
O-B08..B14	UC2_*	—	—	—	—	10 ms	(Identical signals to Unit Controller 2)

3.3.2 Outputs to Gas Turbine

#	Signal Name	Type	Dim	Units	Range	Description
O-B20	GT_Start	Int	1	bool	0/1	Gas turbine start command
O-B21	GT_Stop	Int	1	bool	0/1	Gas turbine stop command
O-B22	GT_P_cmd	Real	1	kW	0 – 500	Gas turbine power setpoint

3.3.3 Outputs to Breakers

#	Signal Name	Type	Dim	Units	Range	Description
O-B25	Utility_BRK_cmd	Int	1	bool	0/1	Utility breaker close/open command
O-B26	Load-Shed_cmd[4]	Int	4	bool	0/1	Load shedding breaker commands (per tier)

3.3.4 Outputs to EMS / SCADA (Layer B → Layer C)

#	Signal Name	Type	Dim	Units	Range	Update Rate	Description
O-B30	P_plant_meas	Real	1	kW	-10000 to +10000	100 ms	Aggregate plant active power
O-B31	Q_plant_meas	Real	1	kVAR	-10000 to +10000	100 ms	Aggregate plant reactive power
O-B32	SOC_avg	Real	1	%	0 – 100	100 ms	Average SOC across units
O-B33	SOC_min	Real	1	%	0 – 100	100 ms	Minimum SOC across units
O-B34	SOC_max	Real	1	%	0 – 100	100 ms	Maximum SOC across units
O-B35	V_bus	Real	1	V	0 – 15000	100 ms	13.2 kV bus voltage
O-B36	f_meas	Real	1	Hz	55 – 65	100 ms	Measured grid frequency
O-B37	MGC_State	Int	1	enum	0–7	100 ms	Current MGC state machine state
O-B38	MGC_Status	Int	1	bitmask	0x0000–0xFFFF	100 ms	Aggregate status word

#	Signal Name	Type	Dim	Units	Range	Update Rate	Description
O-B39	MGC_Fault	Int	1	bitmask	0x0000–0xFFFF	100 ms	Aggregate fault word
O-B40	P_plant_avail	Real	1	kW	0 – 10000	100 ms	Total available discharge power
O-B41	P_plant_chg_avail	Real	1	kW	0 – 10000	100 ms	Total available charge power
O-B42	N_units_online	Int	1	count	0–2	100 ms	Number of units in RUNNING state
O-B43	MGC_heartbeat	Int	1	counter	0–65535	100 ms	Heart-beat to EMS

3.3.5 MGC_Status Bit Definitions

Bit	Name	Description
0	GRID_CONNECTED	Utility breaker closed, grid-tied
1	ISLANDED	Operating in island mode
2	SYNC_READY	Resync conditions met, ready to reconnect
3	BLACK_START_ACTIVE	Black-start sequence in progress
4	LOAD_SHED_ACTIVE	One or more loads shed
5	GT_ONLINE	Gas turbine synchronized and loaded
6	ALL_UNITS_OK	All unit controllers in RUNNING with no faults
7	DR_ACTIVE	Demand response event in progress
8	FREQ_ALARM	Frequency outside 59.5–60.5 Hz
9	VOLT_ALARM	Voltage outside 0.95–1.05 pu
10	SOC_ALARM	Average SOC < 20%
11–15	Reserved	—

3.3.6 MGC_Fault Bit Definitions

Bit	Name	Description	Action
0	UC1_FAULT	Unit Controller 1 has active fault	Redistribute load to UC2
1	UC2_FAULT	Unit Controller 2 has active fault	Redistribute load to UC1
2	BOTH_UC_FAULT	Both unit controllers faulted	Emergency shutdown
3	ISLAND_DETECT	Unintentional island detected	Transition to IS-LANDED state
4	SYNC_FAIL	Resynchronization failed (timeout)	Remain islanded
5	GT_FAULT	Gas turbine fault	Shed GT load to BESS
6	UNDERFREQ	$f < 59.3 \text{ Hz}$ for $> 0.5 \text{ s}$	Load shedding
7	OVERFREQ	$f > 60.5 \text{ Hz}$ for $> 0.5 \text{ s}$	Reduce generation
8	EMS_COMM_LOSS	EMS heartbeat lost ($> 5 \text{ s}$)	Use last valid dispatch
9–15	Reserved	—	—

3.4 MGC Internal Registers

Register	Type	Init	Description
<code>mgc_state</code>	Enum	INIT	Current state machine state
<code>P_total_cmd</code>	Real	0.0	Total plant P command (after sharing)
<code>Q_total_cmd</code>	Real	0.0	Total plant Q command (after sharing)
<code>island_timer</code>	Real	0.0	Island detection confirmation timer
<code>resync_timer</code>	Real	0.0	Resync condition hold timer
<code>blackstart_step</code>	Int	0	Black-start sequence step counter
<code>loadshed_level</code>	Int	0	Current load shedding tier (0=none)
<code>mgc_heartbeat_cnt</code>	Int	0	MGC heartbeat counter
<code>ems_hb_timer</code>	Real	0.0	Time since last EMS heartbeat
<code>last_valid_P_disp</code>	Real	0.0	Last valid P dispatch (for comm loss)
<code>last_valid_Q_disp</code>	Real	0.0	Last valid Q dispatch (for comm loss)

3.5 MGC Control Logic

3.5.1 SOC-Weighted Power Sharing Algorithm

This is the core power distribution algorithm. Power is allocated proportionally to SOC for discharge, and inversely for charge, to equalize SOC across units.

INPUTS:

P_total ☐ total plant P command (from EMS **or** state machine)
 Q_total ☐ total plant Q command
 SOC_1, SOC_2 ☐ unit SOC values
 UC1_Status, UC2_Status ☐ unit status words
 UC1_P_avail, UC2_P_avail ☐ available power per unit

OUTPUTS:

P_cmd_1, P_cmd_2 ☐ individual unit P commands
 Q_cmd_1, Q_cmd_2 ☐ individual unit Q commands

LOGIC:

```

// — Check unit availability —
unit1_ok = (UC1_Status & BIT(READY)) AND NOT (UC1_Fault & HARD_FAULT_MASK)
unit2_ok = (UC2_Status & BIT(READY)) AND NOT (UC2_Fault & HARD_FAULT_MASK)

// — Fallback: single unit operation —
IF NOT unit1_ok AND NOT unit2_ok THEN
  P_cmd_1 = 0; P_cmd_2 = 0
  Q_cmd_1 = 0; Q_cmd_2 = 0
  RETURN

IF NOT unit1_ok THEN
  P_cmd_1 = 0; Q_cmd_1 = 0
  P_cmd_2 = CLAMP(P_total, -P_RATED, +P_RATED)
  Q_cmd_2 = CLAMP(Q_total, -Q_RATED, +Q_RATED)
  RETURN

IF NOT unit2_ok THEN
  P_cmd_2 = 0; Q_cmd_2 = 0
  P_cmd_1 = CLAMP(P_total, -P_RATED, +P_RATED)
  Q_cmd_1 = CLAMP(Q_total, -Q_RATED, +Q_RATED)
  RETURN

// — Both units available: SOC-weighted sharing —
// Prevent division by zero
SOC_sum = MAX(SOC_1 + SOC_2, 1.0)

IF P_total ☐ 0 THEN
  // DISCHARGE: give more to higher-SOC unit
  weight_1 = SOC_1 / SOC_sum
  weight_2 = SOC_2 / SOC_sum
ELSE
  // CHARGE: give more to lower-SOC unit
  weight_1 = (100.0 ☐ SOC_1) / MAX((200.0 ☐ SOC_1 ☐ SOC_2), 1.0)
  weight_2 = (100.0 ☐ SOC_2) / MAX((200.0 ☐ SOC_1 ☐ SOC_2), 1.0)
END IF

P_cmd_1_raw = P_total ☒ weight_1
P_cmd_2_raw = P_total ☒ weight_2

// — Enforce per-unit limits —
P_cmd_1 = CLAMP(P_cmd_1_raw, -UC1_P_chg_avail, UC1_P_avail)
P_cmd_2 = CLAMP(P_cmd_2_raw, -UC2_P_chg_avail, UC2_P_avail)

// — Redistribute any clipped power —
P_residual = P_total ☐ (P_cmd_1 + P_cmd_2)
IF |P_residual| > 10 THEN // > 10 kW tolerance
  // Try to assign residual to the unclamped unit
  IF P_cmd_1 == P_cmd_1_raw THEN
    P_cmd_1 = CLAMP(P_cmd_1 + P_residual, -UC1_P_chg_avail, UC1_P_avail)
  
```

```

ELSE IF P_cmd_2 == P_cmd_2_raw THEN
    P_cmd_2 = CLAMP(P_cmd_2 + P_residual, -UC2_P_chg_avail, UC2_P_avail)
END IF
END IF

// — Q sharing: equal split (Q is not SOC-dependent) —
Q_cmd_1 = Q_total / 2.0
Q_cmd_2 = Q_total / 2.0

// Verify apparent power limits per unit
FOR i IN {1, 2}:
    S_i = SQRT(P_cmd_i^2 + Q_cmd_i^2)
    IF S_i > S_RATED THEN
        // Scale Q down to stay within S limit, P takes priority
        Q_max = SQRT(S_RATED^2 - P_cmd_i^2)
        Q_cmd_i = SIGN(Q_cmd_i) * MIN(|Q_cmd_i|, Q_max)
    END IF

```

Example — SOC-weighted sharing:

Scenario	SOC ₁	SOC ₂	P _{total}	P _{cmd1}	P _{cmd2}
Equal SOC	50%	50%	6000 kW	3000 kW	3000 kW
Unequal SOC (discharge)	80%	40%	6000 kW	4000 kW	2000 kW
Unequal SOC (charge)	80%	40%	-6000 kW	-2000 kW	-4000 kW
One unit faulted	50%	FAULT	4000 kW	4000 kW	0 kW
SOC near limit	12%	80%	6000 kW	652 kW	5000 kW

3.5.2 Islanding Detection (Passive Method)

The MGC uses passive detection based on Rate of Change of Frequency (ROCOF) and vector surge.

```

CONSTANTS:
    ROCOF_THRESHOLD == 1.5 Hz/s // Rate of change of frequency threshold
    VECTOR_SURGE_TH == 10.0 deg // Phase angle shift threshold
    ISLAND_CONFIRM_T == 0.1 s // Confirmation time
    FREQ_LOW_TH == 59.3 Hz // Underfrequency alarm
    FREQ_HIGH_TH == 60.5 Hz // Overfrequency alarm

INPUTS:
    f_grid          - measured frequency
    df_dt           - rate of change of frequency
    V_phase_angle   - grid voltage phase angle
    Utility_BRK     - utility breaker status

INTERNAL:
    prev_phase_angle - previous cycle phase angle
    island_timer     - confirmation timer

LOGIC (executed every 10 ms):

// — Direct detection: breaker opened —
IF Utility_BRK == 0 AND prev_Utility_BRK == 1 THEN
    island_detected == TRUE
    RETURN

// — ROCOF detection —
rocof_trip == (df_dt > ROCOF_THRESHOLD)

// — Vector surge detection —
delta_angle == V_phase_angle - prev_phase_angle
// Unwrap angle
IF delta_angle > 180 THEN delta_angle -= 360
IF delta_angle < -180 THEN delta_angle += 360
vector_surge_trip == (delta_angle > VECTOR_SURGE_TH)

// — Combined detection with confirmation —
IF rocof_trip OR vector_surge_trip THEN
    island_timer += Ts_mgc
    IF island_timer > ISLAND_CONFIRM_T THEN
        island_detected == TRUE
    END IF
ELSE
    island_timer == 0
END IF

prev_phase_angle == V_phase_angle
prev_Utility_BRK == Utility_BRK

```

3.5.3 Resynchronization Logic

Before reconnecting to the grid after an island event, the MGC verifies sync conditions.

```

CONSTANTS:
    SYNC_DV_MAX    = 0.03          // 3% voltage difference
    SYNC_DF_MAX    = 0.1 Hz        // Frequency difference
    SYNC_DPHI_MAX  = 10.0 deg      // Phase angle difference
    SYNC_HOLD_TIME = 1.0 s         // Conditions must hold for this duration
    SYNC_TIMEOUT   = 30.0 s        // Maximum time to achieve sync

INPUTS:
    V_grid, V_island    - grid and island voltage magnitudes (pu)
    f_grid, f_island    - grid and island frequencies
    phi_grid, phi_island - grid and island phase angles

LOGIC:

    dV = V_grid - V_island // V_grid
    df = f_grid - f_island
    dphi = phi_grid - phi_island
    // Unwrap dphi
    IF dphi > 180 THEN dphi = 360 - dphi

    sync_ok = (dV < SYNC_DV_MAX) AND (df < SYNC_DF_MAX) AND (dphi < SYNC_DPHI_MAX)

    IF sync_ok THEN
        resync_timer += Ts_mgc
        IF resync_timer > SYNC_HOLD_TIME THEN
            // Conditions stable - close utility breaker
            SYNC_READY = TRUE
            Utility_BRK_cmd = 1
        END IF
    ELSE
        resync_timer = 0
        SYNC_READY = FALSE
    END IF

    // Timeout protection
    IF total_resync_time > SYNC_TIMEOUT THEN
        SYNC_FAIL = TRUE
        // Remain islanded, alert operator
    END IF

```

3.5.4 Load Shedding Controller

Load shedding operates on a 4-tier priority system. The lowest priority loads are shed first.

CONSTANTS:

```
// Tier thresholds (cumulative)
FREQ_SHED_1 = 59.5 Hz    // Shed Tier 4 (lowest priority)
FREQ_SHED_2 = 59.3 Hz    // Shed Tier 3
FREQ_SHED_3 = 59.0 Hz    // Shed Tier 2
FREQ_SHED_4 = 58.5 Hz    // Shed Tier 1 (highest priority) [ ] last resort)

SOC_SHED_1 = 20.0 %      // Begin shedding at 20% avg SOC
SOC_SHED_2 = 15.0 %      // Aggressive shedding

SHED_CONFIRM = 0.5 s      // Confirmation time before shedding
RESTORE_CONFIRM = 5.0 s   // Confirmation time before restoring
```

LOAD PRIORITY TABLE:

Tier	Priority	Example Loads	Shed Order
4	Lowest	HVAC, lighting (non-essential)	First
3	Low	General outlets, elevators	Second
2	Medium	Data center cooling, emergency lighting	Third
1	Highest	Life safety, fire systems, IT core	Last (emergency only)

LOGIC:

```
// [ ] Determine required shed level [ ]
IF f_grid < FREQ_SHED_4 OR SOC_avg < 10 THEN
    required_level = 4 // Shed all possible
ELSE IF f_grid < FREQ_SHED_3 OR SOC_avg < 12 THEN
    required_level = 3
ELSE IF f_grid < FREQ_SHED_2 OR SOC_avg < 15 THEN
    required_level = 2
ELSE IF f_grid < FREQ_SHED_1 OR SOC_avg < SOC_SHED_1 THEN
    required_level = 1
ELSE
    required_level = 0 // No shedding needed

// [ ] Progressive shedding with confirmation [ ]
IF required_level > loadshed_level THEN
    shed_timer += Ts_mgc
    IF shed_timer > SHED_CONFIRM THEN
        loadshed_level = loadshed_level + 1 // Shed one tier at a time
        LoadShed_cmd[5 [ ] loadshed_level] = 0 // Open tier breaker
        shed_timer = 0
    END IF
ELSE IF required_level < loadshed_level THEN
    restore_timer += Ts_mgc
    IF restore_timer > RESTORE_CONFIRM THEN
        LoadShed_cmd[5 [ ] loadshed_level] = 1 // Close tier breaker
        loadshed_level = loadshed_level [ ] 1 // Restore one tier at a time
        restore_timer = 0
    END IF
ELSE
    shed_timer = 0
    restore_timer = 0
END IF
```

3.5.5 Black-Start Sequence

The black-start procedure energizes the microgrid from a fully de-energized state using the BESS.

BLACK-START SEQUENCE (executed **as** step-based state machine):

STEP 0: INIT

Preconditions: All breakers open, at least 1 UC with SOC > 30%

Action: Select lead unit (highest SOC)

➡ STEP 1

STEP 1: LEAD_UNIT_PRECHARGE

Action: Send Enable=1, Mode_cmd=IDLE to lead UC

Wait: UC enters PRECHARGE, DC bus charges to 1332V $\pm 5\%$

Timeout: 10 s

➡ STEP 2 on DC_OK

➡ ABORT on timeout

STEP 2: LEAD_UNIT_GFM

Action: Send Mode_cmd=GFM to lead UC

Wait: Lead UC transitions to RUNNING **in** GFM mode

Action: Close lead unit MCB (energizes 690V bus)

Timeout: 5 s

➡ STEP 3 on MCB confirmed closed **and** V_{ac} stable

➡ ABORT on timeout **or** fault

STEP 3: ENERGIZE_MV_BUS

Action: Close LV➡MV step-up transformer breaker

Wait: 13.2 kV bus energized, V within $\pm 5\%$ of nominal

Timeout: 5 s

➡ STEP 4 on V_{bus} stable

➡ ABORT on fault

STEP 4: START_GAS_TURBINE (optional)

Action: Send GT_Start = 1

Wait: GT_Status = Running AND GT_Ready = 1

Timeout: 120 s (GT startup **is** slow)

➡ STEP 5 on GT ready

➡ STEP 5 on timeout (**continue** without GT)

STEP 5: SYNC_SECOND_UNIT (**if** available)

Action: Enable UC2, Mode_cmd=GFL

Wait: UC2 MCB closed, running **in** GFL mode

Timeout: 10 s

➡ STEP 6 on UC2 running

➡ STEP 6 on timeout (**continue** with single unit)

STEP 6: LOAD_PICKUP

Action: Close priority **load** breakers sequentially

Sequence: Tier 1 ➡ wait 2s ➡ Tier 2 ➡ wait 2s ➡ Tier 3 ➡ wait 2s ➡ Tier 4

Monitor: Frequency **and** voltage after each step

IF $f < 59.0$ Hz OR $V < 0.90$ pu AFTER any **load** pickup:

Halt **load** pickup, wait **for** stabilization

➡ STEP 7 on all loads connected **or** stable

STEP 7: COMPLETE

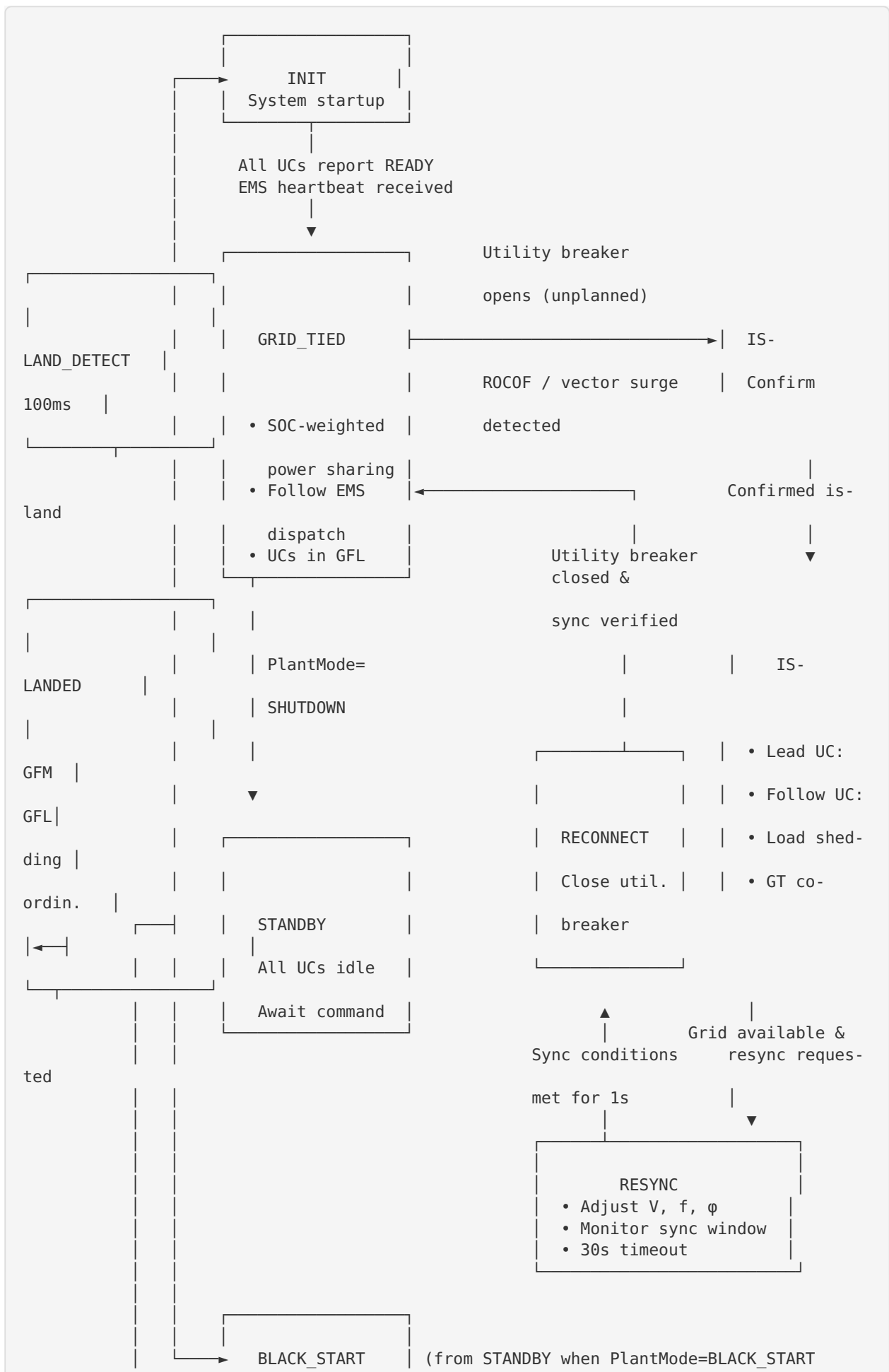
Action: Set MGC_State = ISLANDED (**or** attempt grid resync **if** utility available)

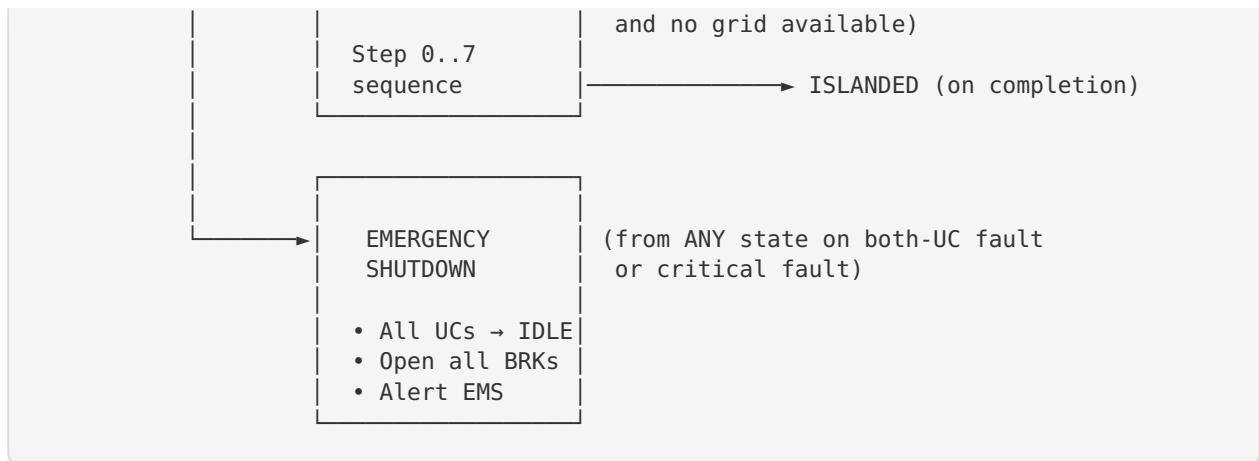
Black-start complete

3.5.6 Plant Mode Definitions

PlantMode	Name	Description	UC Mode	Grid
0	SHUTDOWN	All units idle	IDLE	N/A
1	GRID_TIED_PQ	Normal grid-tied operation	GFL	Connected
2	GRID_TIED_GFM	Grid-tied with droop support	GFM	Connected
3	ISLAND	Islanded operation	GFM (lead), GFL (follow)	Disconnected
4	BLACK_START	Black-start sequence	GFM → GFL	Disconnected
5	MAINTENANCE	One unit at a time for testing	Varies	Connected

3.6 MGC State Machine

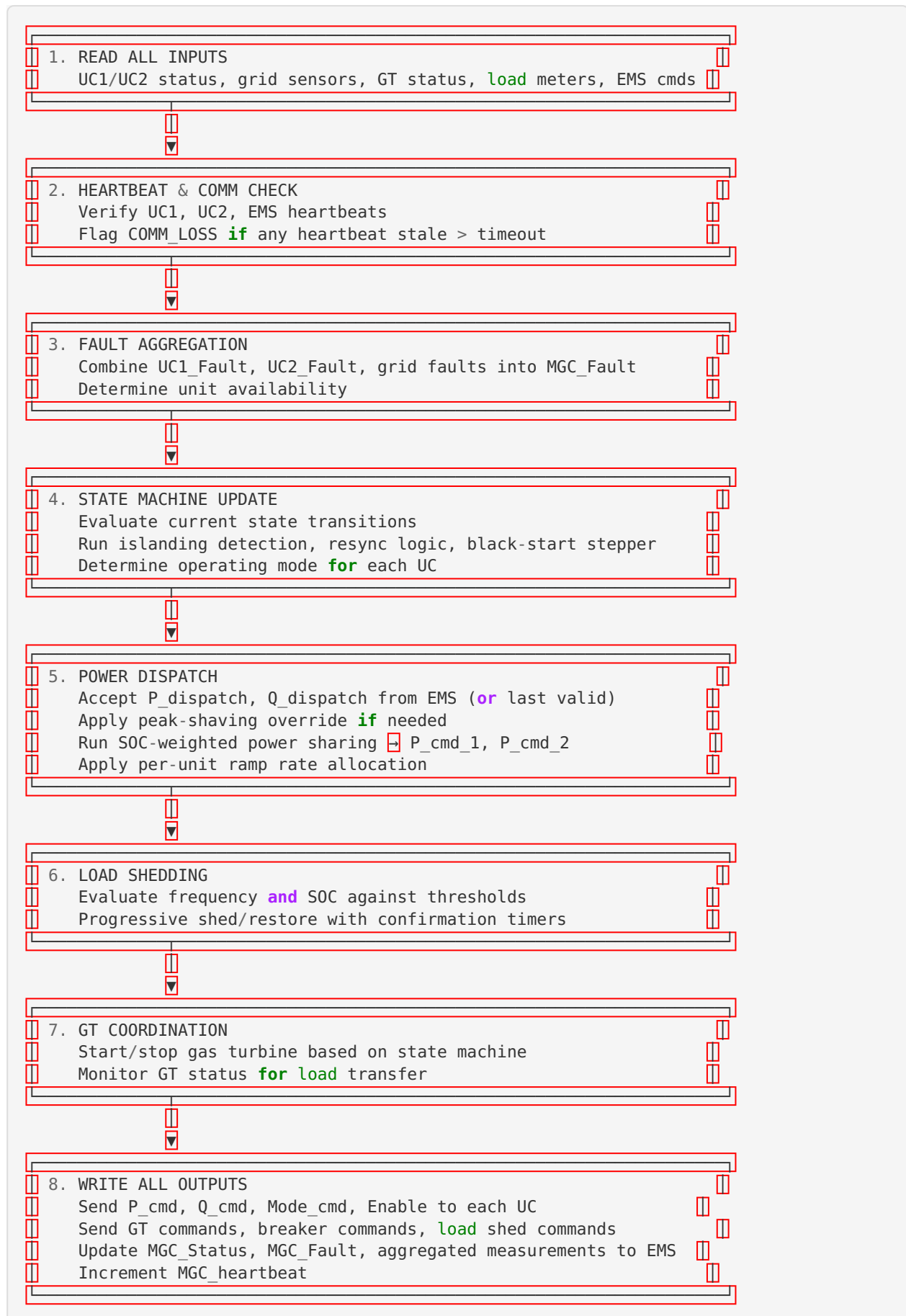




State Transition Table:

From	To	Trigger	Actions
INIT	GRID_TIED	All UCs READY, utility BRK=1, EMS HB valid	Enable UCs in GFL mode
INIT	STANDBY	UCs READY but no grid	Wait for commands
GRID_TIED	ISLAND_DETECT	ROCOF>1.5Hz/s OR vector surge>10° OR utility BRK→0	Start 100ms confirm timer
ISLAND_DETECT	ISLANDED	Timer > 100ms confirmed	Lead UC→GFM, open utility BRK, start GT
ISLAND_DETECT	GRID_TIED	Condition clears within 100ms	Resume normal operation
ISLANDED	RESYNC	Grid available AND operator/EMS requests resync	Begin V/f/φ matching
RESYNC	RECONNECT	$\Delta V < 3\%$, $\Delta f < 0.1\text{Hz}$, $\Delta \phi < 10^\circ$ held for 1s	Close utility breaker
RESYNC	ISLANDED	Sync timeout (30s)	Remain islanded, alert
RECONNECT	GRID_TIED	Utility BRK confirmed closed	All UCs → GFL
STANDBY	BLACK_START	Plant-Mode=BLACK_START AND SOC>30%	Begin step sequence
BLACK_START	ISLANDED	Sequence complete	Normal island operation
ANY	EMERGENCY_SHUTDOWN	Both UCs faulted OR critical system fault	All idle, all BRKs open

3.7 MGC Processing Pipeline (per 10 ms cycle)



4. Layer C — SCADA / EMS Dispatch

4.1 Purpose and Scope

The SCADA/EMS layer provides plant-level scheduling and optimization. In the PSCAD simulation, this is implemented as a **dispatch emulator** using lookup tables and simple logic. In a real deployment, this layer would interface with a utility EMS, VPP aggregator, or market platform.

4.2 Input Signal Table

#	Signal Name	Type	Dim	Units	Range	Source	Update Rate	De-scription
I-C01	P_plant_meas	Real	1	kW	-10000 to +10000	MGC	100 ms	Plant aggregate active power
I-C02	Q_plant_meas	Real	1	kVAR	-10000 to +10000	MGC	100 ms	Plant aggregate reactive power
I-C03	SOC_avg	Real	1	%	0 – 100	MGC	100 ms	Average SOC
I-C04	SOC_min	Real	1	%	0 – 100	MGC	100 ms	Minimum SOC across units
I-C05	SOC_max	Real	1	%	0 – 100	MGC	100 ms	Maximum SOC across units
I-C06	V_bus	Real	1	V	0 – 15000	MGC	100 ms	Bus voltage
I-C07	f_meas	Real	1	Hz	55 – 65	MGC	100 ms	Grid frequency
I-C08	MGC_State	Int	1	enum	0–7	MGC	100 ms	MGC state machine state
I-C09	MGC_Status	Int	1	bitmask	0x0000 –0xFFFF	MGC	100 ms	MGC status word
I-C10	MGC_Fault	Int	1	bitmask	0x0000 –0xFFFF	MGC	100 ms	MGC fault word

#	Signal Name	Type	Dim	Units	Range	Source	Update Rate	Description
I-C11	P_plant_avail	Real	1	kW	0 – 10000	MGC	100 ms	Available discharge power
I-C12	P_plant_chg_avail	Real	1	kW	0 – 10000	MGC	100 ms	Available charge power
I-C13	N_units_on-line	Int	1	count	0–2	MGC	100 ms	Online unit count
I-C14	MGC_heartbeat	Int	1	counter	0–65535	MGC	100 ms	MGC heart-beat
I-C15	Load_P_total	Real	1	kW	0 – 10000	Load meter	1 s	Building load
I-C16	TOU_period	Int	1	enum	0–2	Clock/schedule	1 s	0=Off-peak, 1=Mid-peak, 2=On-peak
I-C17	VPP_signal	Real	1	kW	-10000 to +10000	External	Event	VPP dispatch override
I-C18	VPP_active	Int	1	bool	0/1	External	Event	VPP event active
I-C19	Elec_price	Real	1	\$/kWh	0 – 1.0	Market	1 s	Current electricity price
I-C20	Sim_time	Real	1	s	0 – ∞	PSCAD	Δt	

#	Signal Name	Type	Dim	Units	Range	Source	Update Rate	Description
								Simulation time

4.3 Output Signal Table

#	Signal Name	Type	Dim	Units	Range	Update Rate	Description
O-C01	P_dispatch	Real	1	kW	-10000 to +10000	1 s	Active power dispatch to MGC
O-C02	Q_dispatch	Real	1	kVAR	-10000 to +10000	1 s	Reactive power dispatch to MGC
O-C03	Plant-Mode	Int	1	enum	0-5	1 s	Plant mode command
O-C04	DR_flag	Int	1	bool	0/1	1 s	Demand response event flag
O-C05	EMS_heartbeat	Int	1	counter	0-65535	1 s	Heart-beat to MGC
O-C06	Schedule_active	Int	1	bool	0/1	1 s	Schedule is actively dispatching
O-C07	Peak_shave_active	Int	1	bool	0/1	1 s	Peak shaving is active

4.4 EMS Control Logic

4.4.1 Peak-Shaving Algorithm

The peak-shaving algorithm limits utility import power by discharging the BESS when building load exceeds a threshold.

```

CONSTANTS:
    PEAK_THRESHOLD      = 4000 kW      // Import power cap (from utility)
    PEAK_DEADBAND       = 200 kW      // Hysteresis band
    MIN_SOC_PEAK_SHAVE  = 25.0 %      // Don't peak-shave below this SOC
    PEAK_SHAVE_KP       = 1.0         // Proportional gain
    PEAK_SHAVE_KI       = 0.1         // Integral gain (per second)
    PEAK_SHAVE_MAX      = 10000 kW    // Maximum discharge for peak-shaving

INTERNAL:
    ps_integral = 0.0                // Integrator state
    ps_active   = FALSE              // Peak-shaving active flag

LOGIC (executed every 1 s):

    // Net load seen by utility (positive = importing)
    P_utility_import = Load_P_total - P_plant_meas // (-P_plant when discharging)

    // Activation with hysteresis
    IF P_utility_import > PEAK_THRESHOLD + PEAK_DEADBAND THEN
        ps_active = TRUE
    ELSE IF P_utility_import < PEAK_THRESHOLD - PEAK_DEADBAND THEN
        ps_active = FALSE
        ps_integral = 0

    // PI controller to regulate import at threshold
    IF ps_active AND SOC_avg > MIN_SOC_PEAK_SHAVE THEN
        error = P_utility_import - PEAK_THRESHOLD
        ps_integral += error * Ts_ems // Ts_ems = 1.0 s
        ps_integral = CLAMP(ps_integral, 0, PEAK_SHAVE_MAX)

        P_peak_shave = PEAK_SHAVE_KP * error + PEAK_SHAVE_KI * ps_integral
        P_peak_shave = CLAMP(P_peak_shave, 0, PEAK_SHAVE_MAX)
        P_peak_shave = MIN(P_peak_shave, P_plant_avail) // Can't exceed available

        Peak_shave_active = 1
    ELSE
        P_peak_shave = 0
        Peak_shave_active = 0
    END IF

```

4.4.2 TOU Optimization Algorithm

The Time-of-Use optimizer charges during off-peak and discharges during on-peak periods.

```

CONSTANTS:
// TOU Schedule (representative)
// Period 0: Off-peak    (22:00 - 07:00) - low price - CHARGE
// Period 1: Mid-peak    (07:00 - 14:00, 20:00 - 22:00) - medium price - HOLD
// Period 2: On-peak     (14:00 - 20:00) - high price - DISCHARGE

TOU_CHARGE_RATE      = -5000 kW // Charge at rated power during off-peak
TOU_DISCHARGE_RATE    = 5000 kW // Discharge at rated power during on-peak
TOU_SOC_CHARGE_TARGET = 90.0 %  // Stop charging above this SOC
TOU_SOC_DISCHARGE_MIN = 20.0 %  // Stop discharging below this SOC

LOGIC (executed every 1 s):

CASE TOU_period OF
  0 (OFF_PEAK):
    IF SOC_avg < TOU_SOC_CHARGE_TARGET THEN // Charge
      P_tou = TOU_CHARGE_RATE
    ELSE // Full, stop charging
      P_tou = 0
    END IF

  1 (MID_PEAK): // Hold / standby
    P_tou = 0

  2 (ON_PEAK):
    IF SOC_avg > TOU_SOC_DISCHARGE_MIN THEN // Discharge
      P_tou = TOU_DISCHARGE_RATE
    ELSE // Empty, stop discharging
      P_tou = 0
    END IF
END CASE

```

4.4.3 Demand Response Handler

```

LOGIC:

IF VPP_active == 1 THEN
  // VPP/DR event takes priority over TOU schedule
  P_dr = VPP_signal // Use VPP-directed power
  DR_flag = 1
ELSE
  P_dr = 0
  DR_flag = 0
END IF

```

4.4.4 Dispatch Arbitration (Priority Logic)

Multiple dispatch sources are arbitrated with clear priority ordering:

PRIORITY (highest to lowest):

1. Safety constraints (SOC limits, fault conditions)
2. Demand Response / VPP events
3. Peak-shaving
4. TOU optimization
5. Manual operator override
6. Base schedule

LOGIC (executed every 1 s):

```
// ☐ Safety check ☐
IF MGC_Fault & CRITICAL_MASK THEN
    P_dispatch = 0; Q_dispatch = 0
    PlantMode = SHUTDOWN
    RETURN

// ☐ Demand Response (highest operational priority) ☐
IF VPP_active THEN
    P_dispatch = VPP_signal
    Q_dispatch = 0 // DR typically P-only
    PlantMode = GRID_TIED_PQ
    RETURN

// ☐ Peak-shaving ☐
IF Peak_shave_active THEN
    P_base = P_tou // TOU as base
    P_dispatch = P_base + P_peak_shave // Add peak-shave on top
    Q_dispatch = Q_schedule
    PlantMode = GRID_TIED_PQ
    RETURN

// ☐ TOU schedule ☐
P_dispatch = P_tou
Q_dispatch = Q_schedule // From Q dispatch table
PlantMode = GRID_TIED_PQ

// ☐ Clamp to plant capability ☐
IF P_dispatch > 0 THEN
    P_dispatch = MIN(P_dispatch, P_plant_avail)
ELSE
    P_dispatch = MAX(P_dispatch, -P_plant_chg_avail)
END IF
```

4.5 24-Hour Dispatch Schedule Format

For PSCAD simulation, the EMS schedule is driven by a time-series file:

```
// Format: time(s), P_dispatch(kW), Q_dispatch(kVAR), TOU_period, PlantMode
// Compressed simulation: 1 hour = 1 second of sim time (configurable)
```

```
TIME_SCALE = 3600    // 1 sim-second = 1 hour
```

```
SCHEDULE FILE (dispatch_schedule.txt):
```

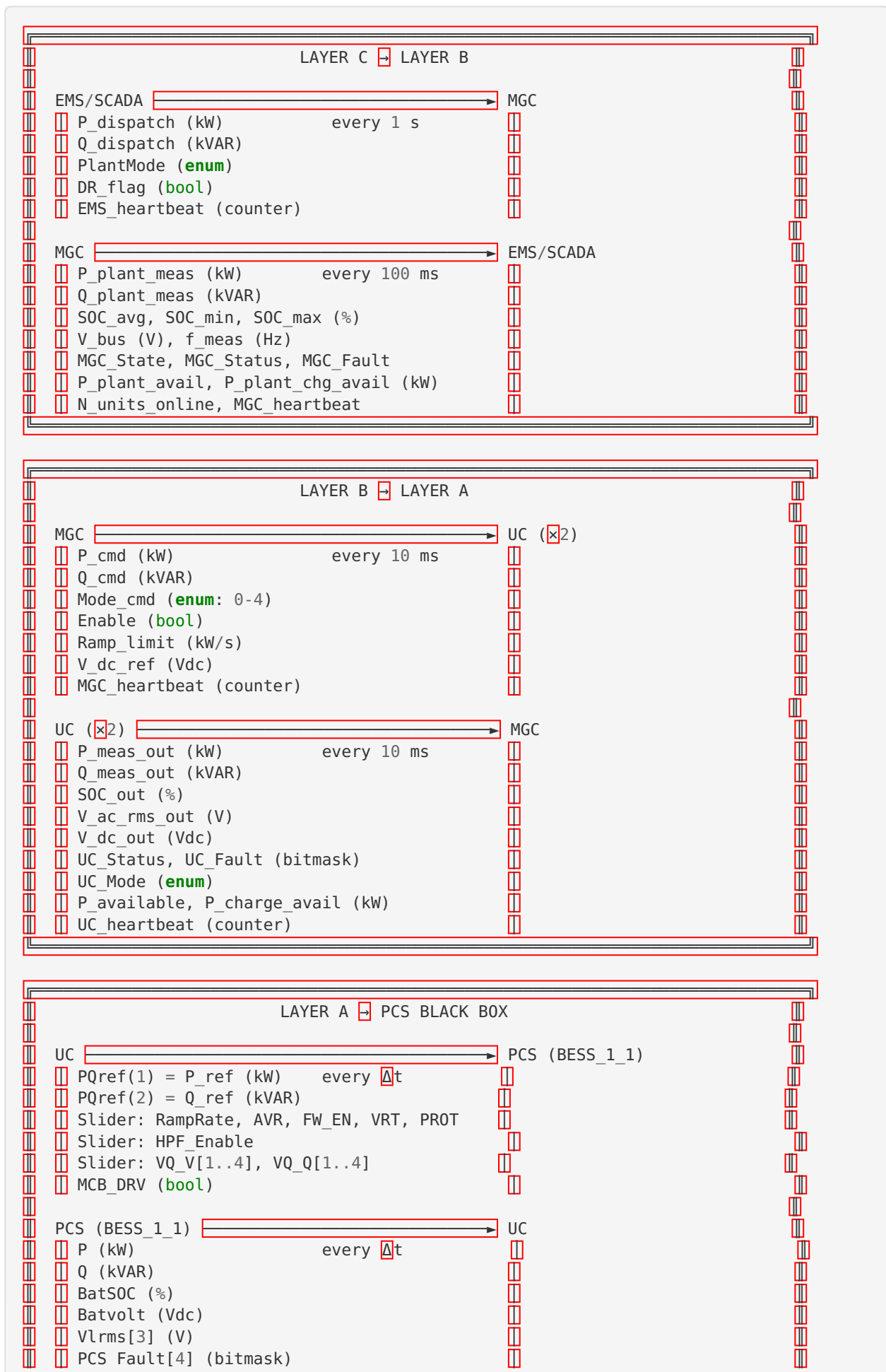
Time(s)	P(kW)	Q(kVAR)	TOU	Mode	Description
0.0	-5000	0	0	1	// 00:00 Off-peak: charge at 5 MW
1.0	-5000	0	0	1	// 01:00 Off-peak: charge
2.0	-5000	0	0	1	// 02:00 Off-peak: charge
3.0	-5000	0	0	1	// 03:00 Off-peak: charge
4.0	-5000	0	0	1	// 04:00 Off-peak: charge
5.0	-3000	0	0	1	// 05:00 Off-peak: reduced charge
6.0	-1000	0	0	1	// 06:00 Off-peak: taper
7.0	0	0	1	1	// 07:00 Mid-peak: standby
8.0	0	0	1	1	// 08:00 Mid-peak: standby
9.0	0	0	1	1	// 09:00 Mid-peak: standby
10.0	0	0	1	1	// 10:00 Mid-peak: standby
11.0	1000	0	1	1	// 11:00 Mid-peak: light discharge
12.0	2000	0	1	1	// 12:00 Mid-peak: moderate discharge
13.0	2000	0	1	1	// 13:00 Mid-peak: moderate discharge
14.0	5000	0	2	1	// 14:00 On-peak: full discharge
15.0	5000	0	2	1	// 15:00 On-peak: full discharge
16.0	5000	0	2	1	// 16:00 On-peak: full discharge
17.0	5000	500	2	1	// 17:00 On-peak: full discharge + Q
18.0	5000	500	2	1	// 18:00 On-peak: full discharge + Q
19.0	3000	0	2	1	// 19:00 On-peak: reduced (SOC low)
20.0	1000	0	1	1	// 20:00 Mid-peak: taper
21.0	0	0	1	1	// 21:00 Mid-peak: standby
22.0	-5000	0	0	1	// 22:00 Off-peak: charge
23.0	-5000	0	0	1	// 23:00 Off-peak: charge

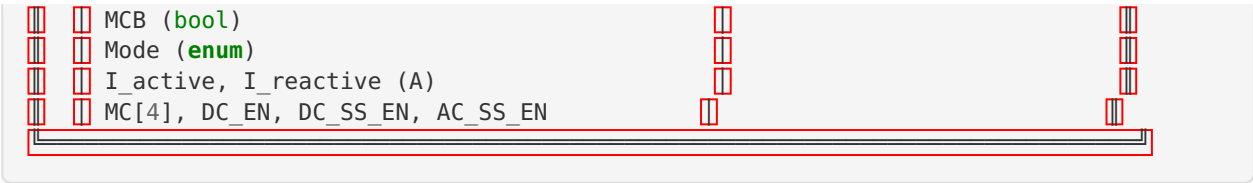
4.6 EMS Processing Pipeline (per 1 s cycle)



5. Inter-Layer Signal Flow

5.1 Complete Signal Flow Diagram





5.2 Signal Count Summary

Interface	Direction	Signal Count	Update Rate
EMS → MGC	Downlink	5 signals	1 s
MGC → EMS	Uplink	14 signals	100 ms
MGC → UC (×2)	Downlink	7 signals × 2 = 14	10 ms
UC → MGC (×2)	Uplink	11 signals × 2 = 22	10 ms
UC → PCS (×2)	Downlink	14 signals × 2 = 28	Δt (50 μs)
PCS → UC (×2)	Uplink	13+ signals × 2 = 26+	Δt (50 μs)
Total inter-layer signals		~109	

5.3 Data Label Naming Convention (PSCAD)

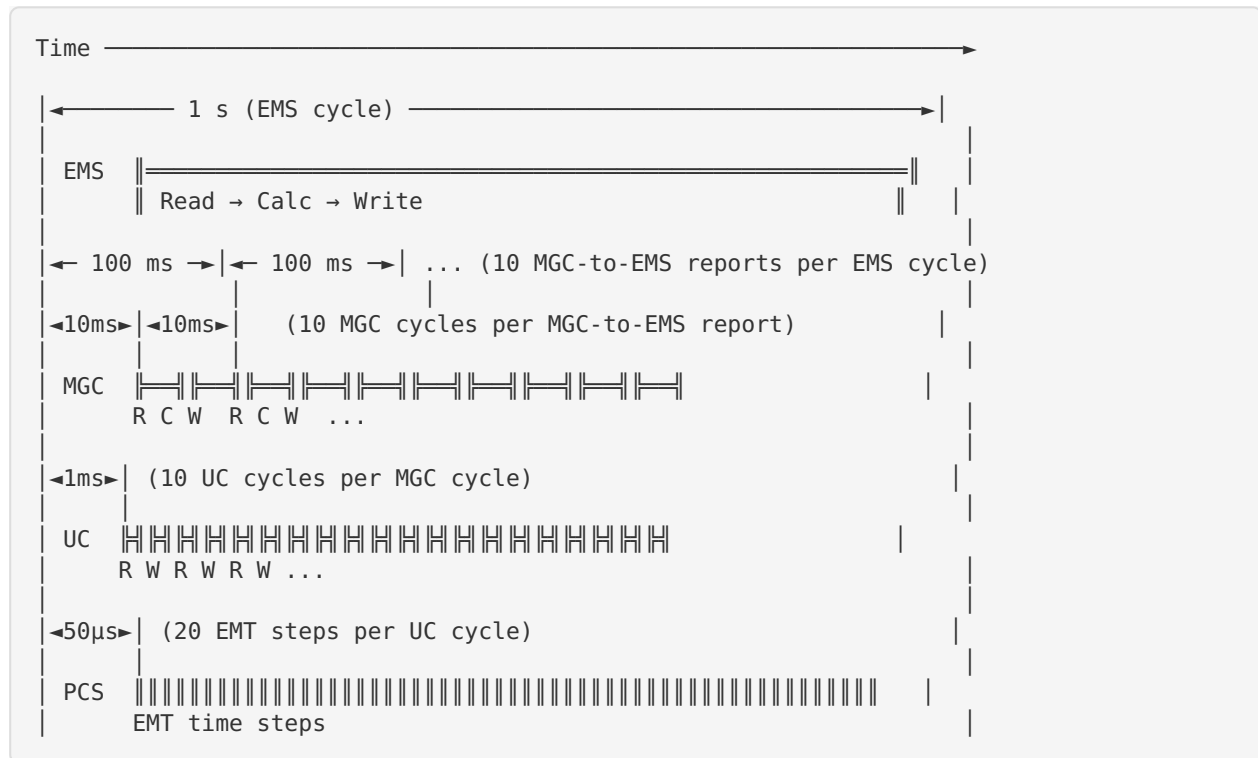
All cross-page signals use a systematic naming convention:

Format: <SOURCE>_<DESTINATION>_<SIGNAL_NAME>

Examples:
EMS_MGC_P_dispatch // EMS to MGC: P dispatch command
MGC_UC1_P_cmd // MGC to UC1: P command
MGC_UC2_P_cmd // MGC to UC2: P command
UC1_MGC_SOC // UC1 to MGC: SOC feedback
UC1_PCS1_PQref // UC1 to PCS1: PQ reference (dim=2)
PCS1_UC1_P_meas // PCS1 to UC1: measured P
MGC_EMS_P_plant // MGC to EMS: plant P measurement

6. Timing Architecture & Sample Rates

6.1 Timing Hierarchy



6.2 Sample Rate Requirements

Controller	Cycle Time	Bandwidth Target	Justification
PCS Internal (Ctrl_1_2)	50 µs (Δt)	500–1000 Hz (current loop)	Requires EMT-rate for dq current control, PWM
Unit Controller	1 ms	10–50 Hz (P/Q outer loop)	Ramp limiter, SOC clamp, mode switching
Microgrid Controller	10 ms	1–10 Hz (droop, sharing)	Power sharing, islanding detection, state machine
MGC → EMS report	100 ms	—	Status reporting, not control-critical
EMS / SCADA	1 s	0.01–0.1 Hz (scheduling)	Dispatch scheduling, optimization

6.3 Latency Budget

Event	Detection	Decision	Action	Total Budget
PCS hardware fault	50 μ s (PCS internal)	1 ms (UC)	1 ms (UC)	< 2 ms
SOC limit reached	1 ms (UC reads SOC)	1 ms (UC clamp logic)	50 μ s (PQref update)	< 3 ms
Island detection	10 ms (MGC RO-COF calc)	100 ms (confirmation)	10 ms (mode switch)	< 120 ms
Load step (20→90%)	50 μ s (PCS current loop)	—	—	< 16 ms (1 cycle @ 60Hz)
Mode transition	1 ms (UC)	10 ms (MGC coordination)	1 ms (UC slider change)	< 4 ms (target)
EMS dispatch change	1 s (EMS cycle)	10 ms (MGC sharing)	1 ms (UC ramp start)	< 1.02 s
Resync completion	10 ms (sync check)	1 s (hold time)	10 ms (breaker close)	$\approx$ 1 s

6.4 PSCAD Multi-Rate Implementation

PSCAD Multi-Rate Configuration:

EMT Simulation Engine $\Delta t = 50 \mu s$ (base time step) All electrical components run at Δt
UC Page Module Execution rate: $\Delta t \times 20 = 1 \text{ ms}$ PSCAD setting: Multi-Rate Factor = 20 Uses: Sampler/Comparator blocks
MGC Page Module Execution rate: $\Delta t \times 200 = 10 \text{ ms}$ PSCAD setting: Multi-Rate Factor = 200 Uses: Sampler/Comparator blocks
EMS Page Module Execution rate: $\Delta t \times 20000 = 1 \text{ s}$ PSCAD setting: Multi-Rate Factor = 20000 Uses: File Reference, Lookup Table

7. Control Algorithm Specifications

7.1 Algorithm Summary Table

ID	Algorithm	Layer	Update Rate	Inputs	Outputs	Section
ALG-01	Ramp-Rate Limiter	UC	1 ms	P_cmd, Ramp_limit	P_ref_inter nal	§2.5.1
ALG-02	SOC Protection Clamp	UC	1 ms	SOC, P_target_raw	P_target	§2.5.2
ALG-03	Fault Handler	UC	1 ms	PCS_Fault, V_dc, SOC, HB	fault_latch	§2.5.3
ALG-04	Mode Switch	UC	1 ms	Mode_cmd	Slider configs	§2.5.4
ALG-05	Available Power Calc	UC	1 ms	SOC	P_available , P_chg_avai l	§2.5.5
ALG-06	SOC-Weighted Sharing	MGC	10 ms	P_total, SOC_1, SOC_2	P_cmd_1, P_cmd_2	§3.5.1
ALG-07	Islanding Detection	MGC	10 ms	f, df/dt, V_angle	is-land_detected	§3.5.2
ALG-08	Resync Logic	MGC	10 ms	V, f, ϕ (grid vs island)	Utility_BRK_cmd	§3.5.3
ALG-09	Load Shedding	MGC	10 ms	f, SOC_avg	Load-Shed_cmd[4]	§3.5.4
ALG-10	Black-Start Sequencer	MGC	10 ms	UC/GT status	Step commands	§3.5.5
ALG-11	Peak-Shaving PI	EMS	1 s	Load_P, P_plant	P_peak_shave	§4.4.1
ALG-12	TOU Optimizer	EMS	1 s	TOU_period, SOC_avg	P_tou	§4.4.2
ALG-13	DR Handler	EMS	1 s			§4.4.3

ID	Algorithm	Layer	Update Rate	Inputs	Outputs	Section
				VPP_signal, VPP_active	P_dr, DR_flag	
ALG-14	Dispatch Arbitration	EMS	1 s	All dis- patch sources	P_dispatch	§4.4.4

7.2 Tuning Parameters Summary

Parameter	Value	Layer	Description
DEFAULT_RAMP	833.33 kW/s	UC	Default ramp rate per unit
SOC_FLOOR	10%	UC	Hard discharge cutoff
SOC_CEILING	95%	UC	Hard charge cutoff
SOC_LOW_WARN	15%	UC	Discharge derating begins
SOC_HIGH_WARN	90%	UC	Charge derating begins
HB_TIMEOUT	500 ms	UC	MGC heartbeat timeout
FAULT_CLEAR_DELAY	5.0 s	UC	Fault clear cooldown
ROCOF_THRESHOLD	1.5 Hz/s	MGC	Islanding detection ROCOF
VECTOR_SURGE_TH	10.0°	MGC	Islanding detection phase
ISLAND_CONFIRM_T	100 ms	MGC	Island detection confirm
SYNC_DV_MAX	3%	MGC	Resync voltage tolerance
SYNC_DF_MAX	0.1 Hz	MGC	Resync frequency tolerance
SYNC_DPHI_MAX	10°	MGC	Resync phase tolerance
SYNC_HOLD_TIME	1.0 s	MGC	Resync condition hold time
SYNC_TIMEOUT	30.0 s	MGC	Resync timeout
FREQ_SHED_1	59.5 Hz	MGC	Load shed tier 4 trigger
FREQ_SHED_2	59.3 Hz	MGC	Load shed tier 3 trigger
FREQ_SHED_3	59.0 Hz	MGC	

Parameter	Value	Layer	Description
			Load shed tier 2 trigger
FREQ_SHED_4	58.5 Hz	MGC	Load shed tier 1 trigger
SOC_SHED_1	20%	MGC	SOC-based shed trigger
PEAK_THRESHOLD	4000 kW	EMS	Peak-shaving import cap
PEAK_SHAVE_KP	1.0	EMS	Peak-shave proportional gain
PEAK_SHAVE_KI	0.1	EMS	Peak-shave integral gain
TOU_CHARGE_RATE	-5000 kW	EMS	Off-peak charge rate
TOU_DISCHARGE_RATE	5000 kW	EMS	On-peak discharge rate

8. State Machine Specifications

8.1 UC State Machine — Formal Definition

STATES = {IDLE, PRECHARGE, CONNECTING, RUNNING, DISCONNECTING, FAULTED, FAULT_CLEAR}

INITIAL_STATE = IDLE

TRANSITIONS:

(IDLE, Enable=1 $\wedge$ Mode_cmd $\in$ {1,2,3} $\wedge$ fault_latch=0) $\rightarrow$ PRECHARGE
 (PRECHARGE, DC_OK=1) $\rightarrow$ CONNECTING
 (PRECHARGE, timeout(10s) $\vee$ fault) $\rightarrow$ FAULTED
 (CONNECTING, MCB_status=1) $\rightarrow$ RUNNING
 (CONNECTING, timeout(5s) $\vee$ fault) $\rightarrow$ FAULTED
 (RUNNING, Enable=0 $\vee$ Mode_cmd=IDLE) $\rightarrow$ DISCONNECTING
 (RUNNING, hard_fault) $\rightarrow$ FAULTED
 (DISCONNECTING, P $\approx$ 0 $\wedge$ MCB=0) $\rightarrow$ IDLE
 (FAULTED, fault_cleared) $\rightarrow$ FAULT_CLEAR
 (FAULT_CLEAR, timer(5s) $\wedge$ Enable=1) $\rightarrow$ CONNECTING
 (FAULT_CLEAR, Enable=0) $\rightarrow$ IDLE

STATE_OUTPUTS:

IDLE: P_ref=0, Q_ref=0, MCB=open
 PRECHARGE: P_ref=0, Q_ref=0, DCC_SS_EN=1
 CONNECTING: P_ref=0, Q_ref=0, MCB=closing, sliders configured
 RUNNING: P_ref=f(P_cmd), Q_ref=f(Q_cmd), MCB=closed
 DISCONNECTING: P_ref $\rightarrow$ 0, Q_ref $\rightarrow$ 0, MCB $\rightarrow$ open (after ramp)
 FAULTED: P_ref=0, Q_ref=0, MCB=open, fault_latch active
 FAULT_CLEAR: P_ref=0, Q_ref=0, timer counting

8.2 MGC State Machine — Formal Definition

```

STATES = {INIT, STANDBY, GRID_TIED, ISLAND_DETECT, ISLANDED,
          RESYNC, RECONNECT, BLACK_START, EMERGENCY_SHUTDOWN}

INITIAL_STATE = INIT

TRANSITIONS:
  (INIT, all_UCs_ready [A] grid_available) [X] GRID_TIED
  (INIT, all_UCs_ready [A] [X]grid_available) [X] STANDBY
  (STANDBY, PlantMode=GRID_TIED [A] grid_available) [X] GRID_TIED
  (STANDBY, PlantMode=BLACK_START [A] SOC>30%) [X] BLACK_START
  (GRID_TIED, island_trigger) [X] ISLAND_DETECT
  (GRID_TIED, PlantMode=SHUTDOWN) [X] STANDBY
  (ISLAND_DETECT, confirmed(100ms)) [X] ISLANDED
  (ISLAND_DETECT, [X]island_trigger within 100ms) [X] GRID_TIED
  (ISLANDED, grid_available [A] resync_requested) [X] RESYNC
  (RESYNC, sync_ok(1s)) [X] RECONNECT
  (RESYNC, timeout(30s)) [X] ISLANDED
  (RECONNECT, utility_BRK=1) [X] GRID_TIED
  (BLACK_START, sequence_complete) [X] ISLANDED
  (ANY, both_UC_fault [V] critical_fault) [X] EMERGENCY_SHUTDOWN
  (EMERGENCY_SHUTDOWN, faults_cleared [A] operator_reset) [X] INIT

STATE_OUTPUTS:
  INIT:           All UCs disabled, monitoring
  STANDBY:        UCs idle, awaiting commands
  GRID_TIED:      SOC-weighted sharing, UCs in GFL, follow EMS
  ISLAND_DETECT:  Confirm timer running, hold current outputs
  ISLANDED:       Lead UC [X]GFM, follower [X]GFL, load shedding active
  RESYNC:         V/f/[X] matching, sync check running
  RECONNECT:      Close utility breaker, transition UCs to GFL
  BLACK_START:    Step-based sequence (§3.5.5)
  EMERGENCY_SHUTDOWN: All UCs [X]IDLE, all breakers open, alert EMS

```

8.3 State Machine Timing Constraints

Transition	Maximum Allowed Time	Notes
IDLE → RUNNING	< 15 s	Precharge (10s) + connect (5s)
GRID_TIED → ISLANDED	< 120 ms	Detection (10ms) + confirm (100ms) + switch (10ms)
ISLANDED → GRID_TIED (re-sync)	< 32 s	Sync hold (1s) + breaker close + margin
BLACK_START → ISLANDED	< 150 s	GT startup dominates (120s)
RUNNING → FAULTED	< 2 ms	Safety-critical path
Any → EMERGENCY_SHUTDOWN	< 10 ms	Immediate upon trigger

9. Protection Coordination

9.1 Protection Layer Hierarchy

Layer 0 (Hardware / PCS Internal) – FASTEST

- └─ IGBT gate blocking on overcurrent ($< 10 \mu\text{s}$)
- └─ SPOV protection (single-pole overvoltage)
- └─ DC bus overvoltage crowbar
- └─ Handled entirely by Ctrl_1_2 – DO NOT MODIFY

Layer 1 (Unit Controller) – FAST

- └─ SOC hard limits (10%, 95%) – 1 ms response
- └─ DC bus voltage limits (1200–1450 V) – 1 ms
- └─ Communication loss detection – 500 ms timeout
- └─ Mode transition safety – 1 ms
- └─ Fault latching and safe shutdown

Layer 2 (Microgrid Controller) – MEDIUM

- └─ Islanding detection (ROCOF + vector surge) – 100 ms
- └─ Load shedding (frequency + SOC based) – 500 ms confirm
- └─ Unit fault redistribution – 10 ms
- └─ Resynchronization safety – 1 s hold time
- └─ Black-start safety interlocks

Layer 3 (EMS/SCADA) – SLOW

- └─ Plant-level operational limits – 1 s
- └─ SOC management (TOU cycling) – minutes
- └─ Operator alarms and notifications

9.2 Protection Coordination Matrix

Fault Condition	Layer 0 (PCS)	Layer 1 (UC)	Layer 2 (MGC)	Layer 3 (EMS)
IGBT short circuit	Gate block <10 μ s	—	—	—
Phase overcurrent	OC blocking	Log fault	Redistribute load	Alarm
DC overvoltage (>1500V)	Crowbar	Reduce charge	—	—
DC overvoltage (>1450V)	—	Block charge	—	—
DC undervoltage (<1200V)	—	Block discharge	—	—
SOC < 10%	—	Block discharge	Start GT, shed load	—
SOC > 95%	—	Block charge	Reduce charge dispatch	—
AC undervoltage (<0.88pu)	VRT ride-through	VRT flag	Island if persistent	—
AC overvoltage (>1.10pu)	VRT ride-through	VRT flag	Island if persistent	—
Frequency < 59.5 Hz	—	—	Shed Tier 4	—
Frequency < 59.0 Hz	—	—	Shed Tiers 2-4	—
Frequency < 58.5 Hz	—	—	Shed all tiers	Emergency
Island detected	—	—	Mode switch, shed	Alarm
Communication loss (UC-MGC)	—	Ramp to 0	Flag unit offline	Alarm
Communication loss (MGC-EMS)	—	—	Use last dispatch	Alarm
	—	—		Critical alarm

Fault Condition	Layer 0 (PCS)	Layer 1 (UC)	Layer 2 (MGC)	Layer 3 (EMS)
Both units faulted			Emergency shut-down	

10. Validation & Test Requirements

10.1 Unit Controller Tests

Test ID	Description	Stimulus	Pass Criteria
UC-T01	Ramp limiter verification	Step P_cmd: 0→5000 kW	P_ref follows ramp at configured rate
UC-T02	SOC floor protection	Discharge at SOC=10%	P_ref clamped to 0
UC-T03	SOC ceiling protection	Charge at SOC=95%	P_ref (charge) clamped to 0
UC-T04	SOC derating (discharge)	Discharge at SOC=12%	P_ref derated to 40% of P_cmd
UC-T05	SOC derating (charge)	Charge at SOC=92%	P_ref (charge) derated to 60% of P_cmd
UC-T06	PCS fault response	Inject PCS_Fault $\neq$ 0	P=Q=0 within 2ms, MCB opens
UC-T07	Heartbeat timeout	Freeze MGC_heartbeat	P ramps to 0 after 500ms
UC-T08	Mode switch GFL→GFM	Mode_cmd: 1→2	AVR=1, FW_EN=1 within 1ms
UC-T09	Apparent power limiting	P=4000, Q=4000	S limited to 5000 kVA
UC-T10	Fault clear and re-start	Clear fault, Enable=1	5s delay, then re-connect

10.2 Microgrid Controller Tests

Test ID	Description	Stimulus	Pass Criteria
MGC-T01	Equal SOC sharing	SOC1=50%, SOC2=50%, P=6MW	P_cmd_1=3MW, P_cmd_2=3MW
MGC-T02	Unequal SOC sharing	SOC1=80%, SOC2=40%, P=6MW	P_cmd_1=4MW, P_cmd_2=2MW
MGC-T03	Single unit fallback	UC1 fault, P=4MW	P_cmd_1=0, P_cmd_2=4MW
MGC-T04	Islanding detection	Open utility BRK	Island detected < 120ms
MGC-T05	ROCOF detection	Inject 2 Hz/s ROCOF	Island detection triggers
MGC-T06	Resynchronization	V, f, ϕ within limits	Utility BRK closed after 1s hold
MGC-T07	Resync timeout	V,f, ϕ never converge	SYNC_FAIL after 30s
MGC-T08	Load shedding - freq	f=59.4 Hz for 0.5s	Tier 4 load shed
MGC-T09	Load shedding - SOC	SOC_avg=18%	Tier 4 load shed
MGC-T10	Black-start sequence	All dead, SOC>30%	Full sequence to IS-LANDED
MGC-T11	Load restore	f recovers to 60 Hz	Loads restored after 5s confirm
MGC-T12	Charge sharing (inverse)	SOC1=80%, SOC2=40%, P=-6MW	P_cmd_1=-2MW, P_cmd_2=-4MW

10.3 EMS/SCADA Tests

Test ID	Description	Stimulus	Pass Criteria
EMS-T01	Peak-shaving activation	Load > 4200 kW	BESS discharges to cap import at 4000 kW
EMS-T02	Peak-shaving deactivation	Load < 3800 kW	Peak-shave ramps down
EMS-T03	TOU off-peak charge	TOU_period=0, SOC<90%	P_dispatch = -5000 kW
EMS-T04	TOU on-peak discharge	TOU_period=2, SOC>20%	P_dispatch = +5000 kW
EMS-T05	TOU SOC limit (full)	TOU_period=0, SOC=92%	P_dispatch = 0 (stop charge)
EMS-T06	DR event override	VPP_active=1	P_dispatch follows VPP_signal
EMS-T07	DR priority over TOU	VPP_active=1 during off-peak	VPP command overrides charge
EMS-T08	Schedule playback	24-hour schedule file	P_dispatch follows schedule
EMS-T09	EMS comm loss	Freeze EMS_heartbeat	MGC uses last valid dispatch
EMS-T10	Safety override	MGC reports critical fault	P_dispatch → 0, Plant-Mode → SHUTDOWN

10.4 Integration Tests

Test ID	Description	Scenario	Pass Criteria
INT-T01	Full dispatch chain	EMS→MGC→UC→PCS	P measured within $\pm 5\%$ of P_dispatch
INT-T02	Utility loss + island + resync	Open utility, wait, re-close	Seamless transition, f $\pm 0.5\text{Hz}$
INT-T03	SOC equalization	Start SOC1=80%, SOC2=40%	SOC converges during cycling
INT-T04	Black start to grid re-connect	Dead bus → black start → resync	Full sequence completes
INT-T05	Cascading load shed + restore	f drops, shed, GT starts, restore	Correct shed/restore sequence
INT-T06	Dual unit fault recovery	Both UC fault → one clears	Single-unit operation resumes
INT-T07	24-hour TOU cycle	Full schedule with peak-shaving	SOC cycles correctly, peak capped

11. PSCAD Implementation Notes

11.1 Page Module Structure

```

DS (Station Definition)
├── Main (page 0)
│   ├── EMS_Dispatch (Layer C – new page module)
│   │   └── Multi-rate factor: 20000 (1 s cycle)
│   ├── MGC_Controller (Layer B – new page module)
│   │   └── Multi-rate factor: 200 (10 ms cycle)
│   ├── UC_1 (Layer A, Unit 1 – new page module)
│   │   └── Multi-rate factor: 20 (1 ms cycle)
│   ├── UC_2 (Layer A, Unit 2 – new page module)
│   │   └── Multi-rate factor: 20 (1 ms cycle)
│   ├── BESS_1_1 (PCS+BESS Unit 1 – existing, unmodified)
│   ├── BESS_1_2 (PCS+BESS Unit 2 – duplicate of BESS_1_1)
│   ├── Source_1 (AC grid equivalent – existing)
│   ├── Multimeter (POI measurements – existing)
│   ├── Gas_Turbine (GT model – new)
│   └── Building_Load (Load model – new)

```

11.2 Signal Routing Method

Method	Use Case	Notes
Data Labels	Cross-page signals (UC↔MGC, MGC↔EMS)	Named with convention from §5.3
Import/Export Tags	High-speed bus signals (UC↔PCS)	Faster than data labels
Direct Wire	Intra-page connections	Within same page module
Slider / Control Panel	Configuration parameters	AuxCtrl sliders for PCS
File Reference	EMS dispatch schedule	dispatch_schedule.txt

11.3 Critical Build Rules

1. **DO NOT** modify any component inside `BESS_1_1` — it is a validated black box
2. **DO NOT** edit auto-generated `.f` files — they are overwritten on rebuild
3. **DO NOT** write `SetupWord` or `CommandWord` integers directly — use slider switches
4. Use Data Labels for all cross-page signal routing
5. Test with `PROT_Enable=0` and `VRT_Enable=0` first, then enable protections
6. UC and MGC pages must use PSCAD multi-rate execution
7. The PCS internal ramp rate (`RampRate` slider) provides a second, independent ramp limit below the UC's external limiter; set to minimum (10 ms) and rely on the UC limiter for coordination
8. Extend simulation duration from 0.5s to ≥ 24 s for full TOU cycle testing (at 1s/hour compression)

11.4 Simulation Time Step Summary

Component	Time Step	Multi-Rate Factor	Notes
EMT Simulation (base)	50 μ s	1	All electrical components
PCS Controller (Ctrl_1_2)	200 μ s	4 (internal)	Set by <code>Tsample</code>
Unit Controller	1 ms	20	New page module
Microgrid Controller	10 ms	200	New page module
EMS Dispatch	1 s	20000	New page module

Appendix A: Signal Index (Alphabetical)

Signal	Layer	Direction	Type	Section
AVR_set	UC→PCS	Output	Int	§2.3.1
DC_EN_status	PCS→UC	Input	Int	§2.2.1
DR_flag	EMS→MGC / MGC→EMS	Both	Int	§4.3, §3.2.5
EMS_heartbeat	EMS→MGC	Output	Int	§4.3
Enable	MGC→UC	Output	Int	§2.2.2
f_grid	Sensor→MGC	Input	Real	§3.2.2
FW_EN_set	UC→PCS	Output	Int	§2.3.1
GT_P_cmd	MGC→GT	Output	Real	§3.3.2
GT_Start/Stop	MGC→GT	Output	Int	§3.3.2
GT_Status	GT→MGC	Input	Int	§3.2.3
I_active	PCS→UC	Input	Real	§2.2.1
Load_P_total	Sensor→MGC/ EMS	Input	Real	§3.2.4, §4.2
LoadShed_cmd[4]	MGC→Breakers	Output	Int	§3.3.3
MCB_DRV	UC→PCS	Output	Int	§2.3.1
MCB_status	PCS→UC	Input	Int	§2.2.1
MGC_heartbeat	MGC→UC/EMS	Output	Int	§3.3.1, §3.3.4
MGC_State	MGC→EMS	Output	Int	§3.3.4
MGC_Status/ Fault	MGC→EMS	Output	Int	§3.3.4
Mode_cmd	MGC→UC	Output	Int	§2.2.2
P_available	UC→MGC	Output	Real	§2.3.2
P_cmd	MGC→UC	Output	Real	§2.2.2
P_dispatch	EMS→MGC	Output	Real	§4.3
P_meas	PCS→UC	Input	Real	§2.2.1

Signal	Layer	Direction	Type	Section
P_plant_meas	MGC→EMS	Output	Real	§3.3.4
P_ref	UC→PCS	Output	Real	§2.3.1
PCS_Fault[4]	PCS→UC	Input	Int	§2.2.1
PlantMode	EMS→MGC	Output	Int	§4.3
Q_cmd	MGC→UC	Output	Real	§2.2.2
Q_dispatch	EMS→MGC	Output	Real	§4.3
Q_ref	UC→PCS	Output	Real	§2.3.1
Ramp_limit	MGC→UC	Output	Real	§2.2.2
SOC	PCS→UC	Input	Real	§2.2.1
SOC_avg/min/max	MGC→EMS	Output	Real	§3.3.4
TOU_period	Clock→EMS	Input	Int	§4.2
UC_Fault	UC→MGC	Output	Int	§2.3.2
UC_heartbeat	UC→MGC	Output	Int	§2.3.2
UC_Mode	UC→MGC	Output	Int	§2.3.2
UC_Status	UC→MGC	Output	Int	§2.3.2
Utility_BRK	Sensor→MGC	Input	Int	§3.2.2
Utility_BRK_cmd	MGC→Breaker	Output	Int	§3.3.3
V_ac_rms[3]	PCS→UC	Input	Real	§2.2.1
V_bat	PCS→UC	Input	Real	§2.2.1
V_dc	PCS→UC	Input	Real	§2.2.1
V_dc_ref	MGC→UC	Output	Real	§2.2.2
V_poi	Sensor→MGC	Input	Real	§3.2.2
VPP_signal	External→EMS	Input	Real	§4.2
VQ_V/Q[4]	UC→PCS	Output	Real	§2.3.1
VRT_Enable_set	UC→PCS	Output	Int	§2.3.1



Appendix B: Acronyms

Acronym	Definition
AVR	Automatic Voltage Regulator (Q-V droop mode)
BESS	Battery Energy Storage System
BRK	Breaker
DR	Demand Response
EMS	Energy Management System
EMT	Electromagnetic Transient
FFT	Fast Fourier Transform
GFL	Grid-Following
GFM	Grid-Forming
GT	Gas Turbine
HPF	High-Pass Filter
HVRT	High-Voltage Ride-Through
LCL	Inductor-Capacitor-Inductor (filter topology)
LVRT	Low-Voltage Ride-Through
MCB	Main Circuit Breaker
MGC	Microgrid Controller
OC	Overcurrent
PCC	Point of Common Coupling
PCS	Power Conversion System
PLL	Phase-Locked Loop
POI	Point of Interconnection
PWM	Pulse-Width Modulation
ROCOF	Rate of Change of Frequency
SCADA	Supervisory Control and Data Acquisition

Acronym	Definition
SOC	State of Charge
SPOV	Single-Pole Overvoltage
SPWM	Sinusoidal Pulse-Width Modulation
SRF	Synchronous Reference Frame
TOU	Time of Use
UC	Unit Controller
VPP	Virtual Power Plant
VRT	Voltage Ride-Through
VSM	Virtual Synchronous Machine

End of Control Architecture Design Document